

Fiber-Reinforced Concrete for Optimized Construction of Light Rail Track Slabs

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ACI Fall Convention , 2021

Wed, October 20th 10:00 AM - 10:20 AM

[Fiber-Reinforced Concrete for Optimized and Sustainable Construction](#)

Development of Alternative Transportation Systems in one of the fastest growing regions of the Southwestern US

Newsire, Oct 19th, 2021

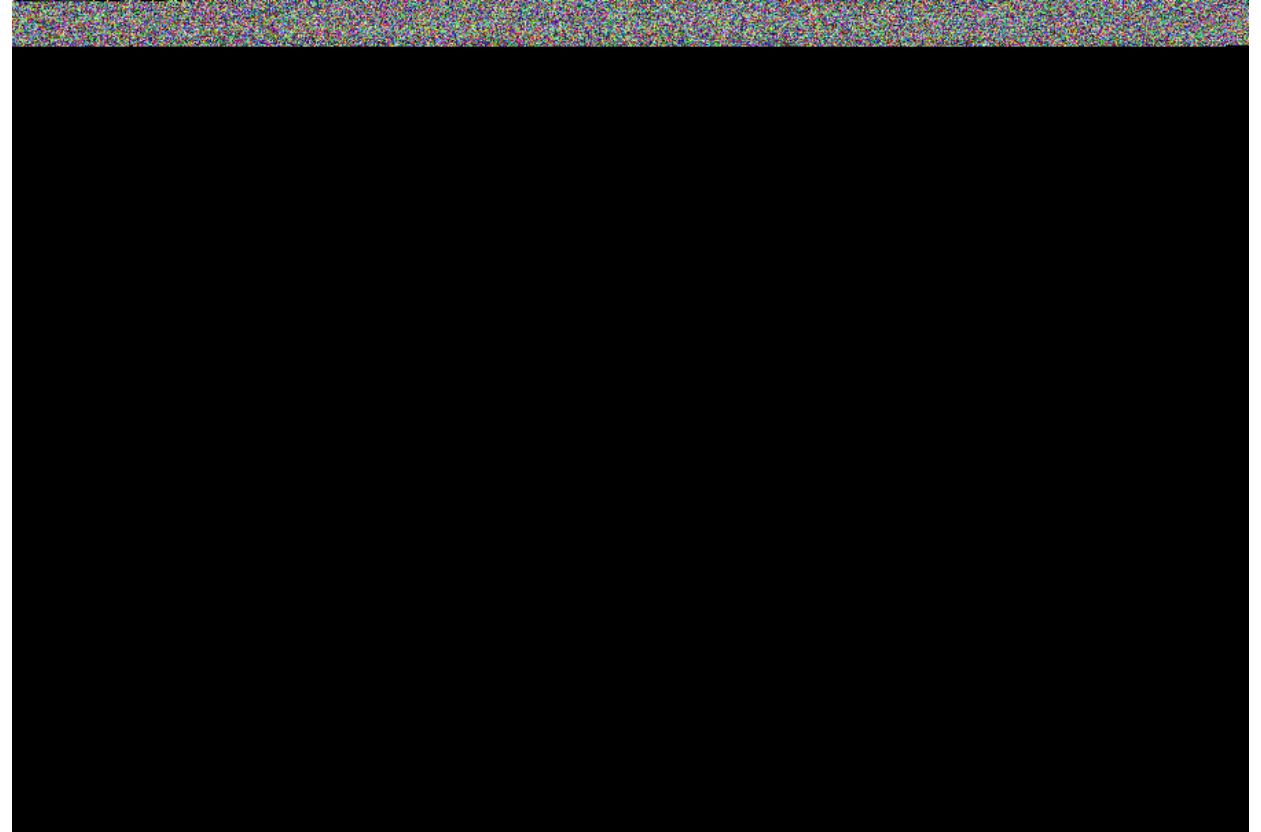
The Howard Hughes Corp. (NYSE: HHC) is partnering to develop Douglas Ranch, a large-scale, master-planned community in Phoenix by purchasing 37,000 acres for the project in Phoenix's West Valley region for \$600 million.

Number of homes: 100,000

Residents: 300,000

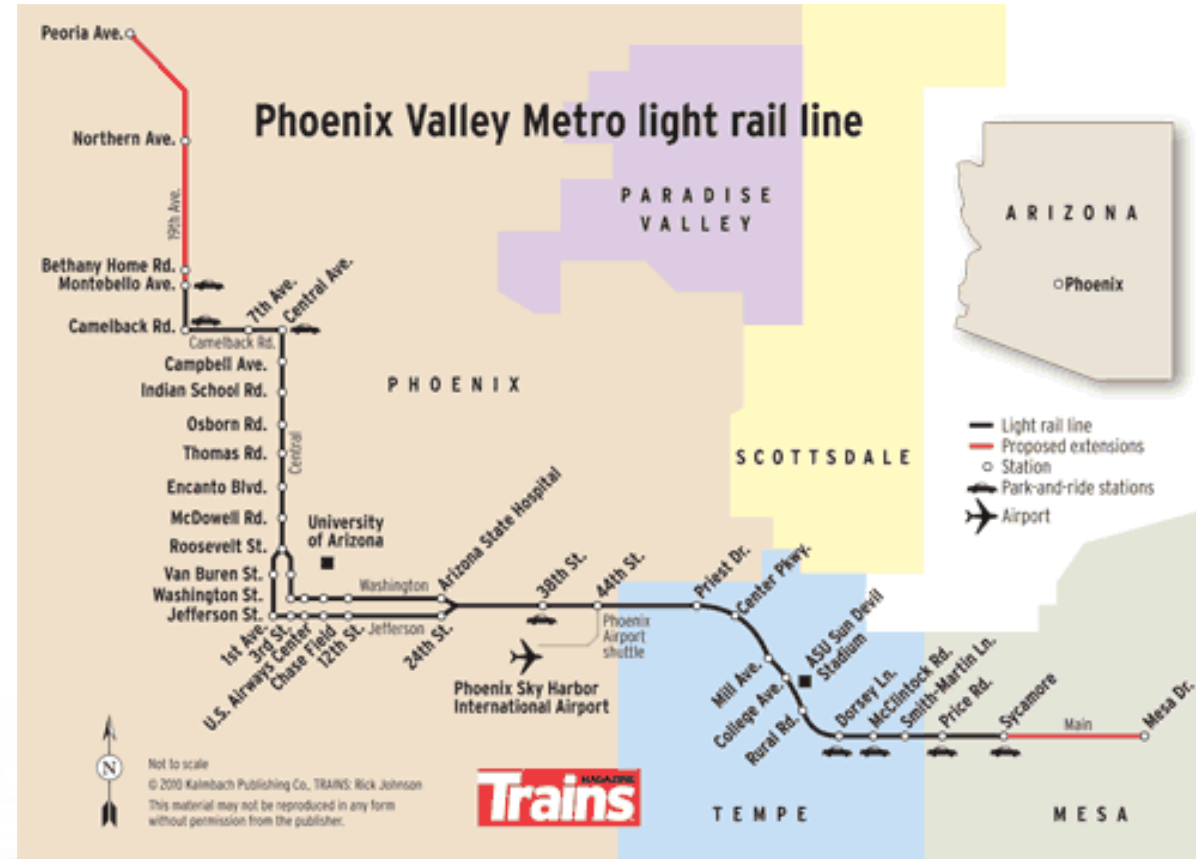
Commercial development: 55 million square feet

Residential lot sales: first half of 2022.



Valley Metro Regional Public Transportation Authority, valleymetro.org

- Unified public brand of regional transit system in the Phoenix, AZ, metropolitan area, responsible for public transit.
- Founded: 1993
- System length: 26.3 miles (42.3 km) (light rail)
- Number of lines: Light rail: 1; Express: 6; Bus: 102
- Annual ridership: 65,958,074
- Daily ridership: 219,406 (3/2017; avg. weekday boardings)



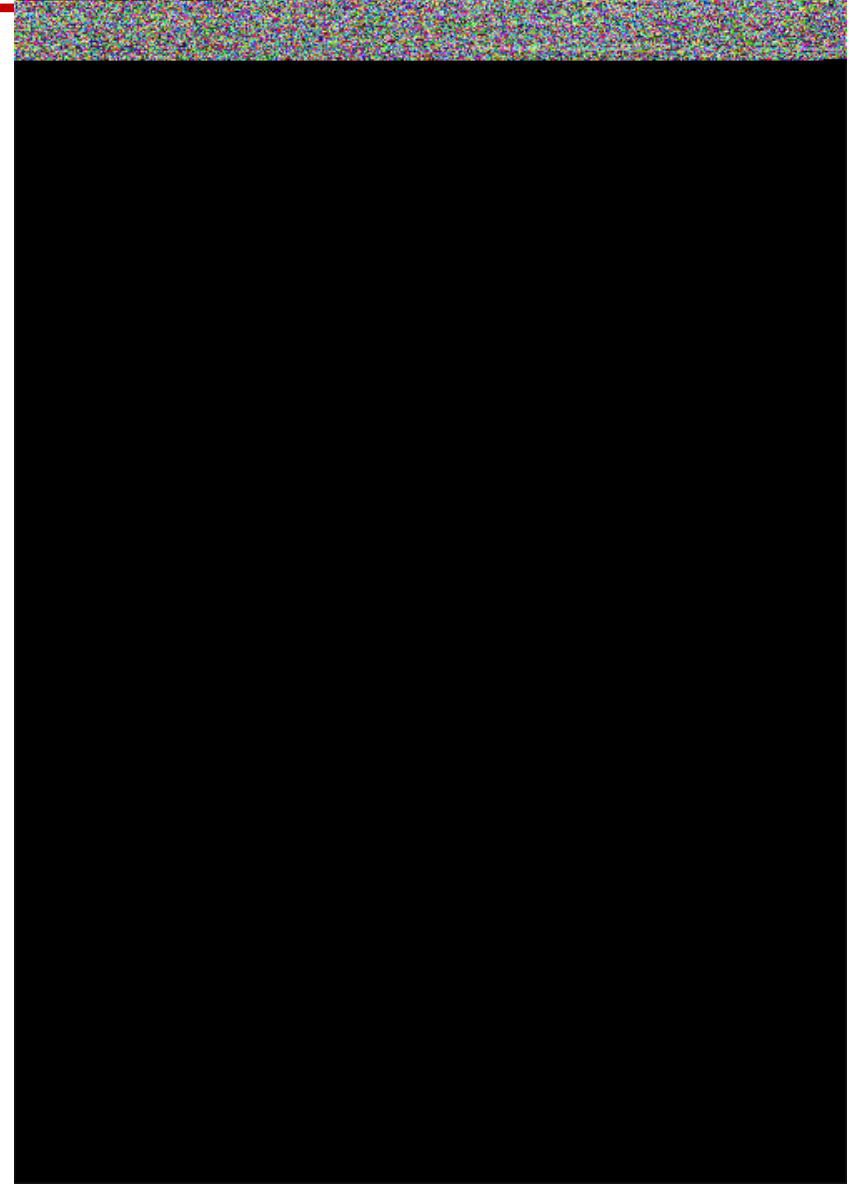
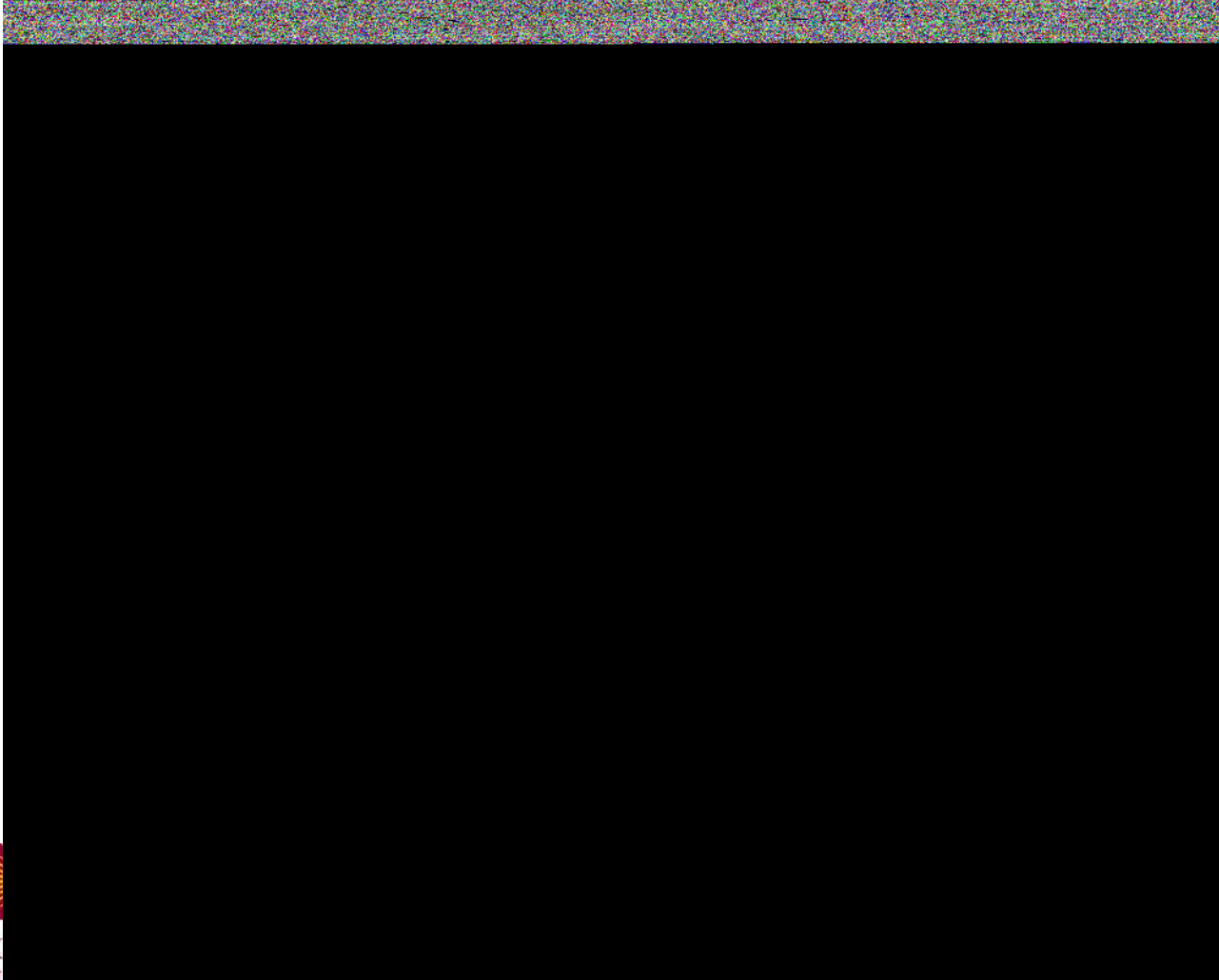
Fiscal Years 2017-2021 Annual Ridership



Valley Metro light rail



City of Phoenix South Central Extension project



Phoenix, Arizona, Proposition 105, End Light Rail Expansion Initiative (August 2019)

- A **yes** vote was a vote in favor of amending the city charter to end construction of light rail extensions; to redirect funds from light rail projects to other transportation infrastructure improvements in Phoenix; and to prohibit funding other light rail development, with an exception for PHX Sky Train
- A **no** vote was a vote against amending the city charter, thereby leaving funds allocated to light rail expansion according to existing charter provisions and city code.
- Prohibited the city from spending money on development, construction, expansion, or improvement of light rail transit, with an exception for PHX Sky Train.
- It would have allocated any revenue from the city's 0.7% transportation sales tax that was previously allocated toward light rail development to other city infrastructure.
- The initiative would have earmarked any revenue allocated to light rail development along Central Avenue south of Washington Street specifically for infrastructure in South Phoenix.
- This would have included [the South Central Extension](#) project. Proposition 105's provisions reallocating revenue would have applied to any collected, unspent revenue as well as future revenue.



Result	Votes	Percentage
Yes	72,752	37.14%
No	123,135	62.86%

Project Overview: Light Rail, Phoenix (Valley Metro and the City of Phoenix)



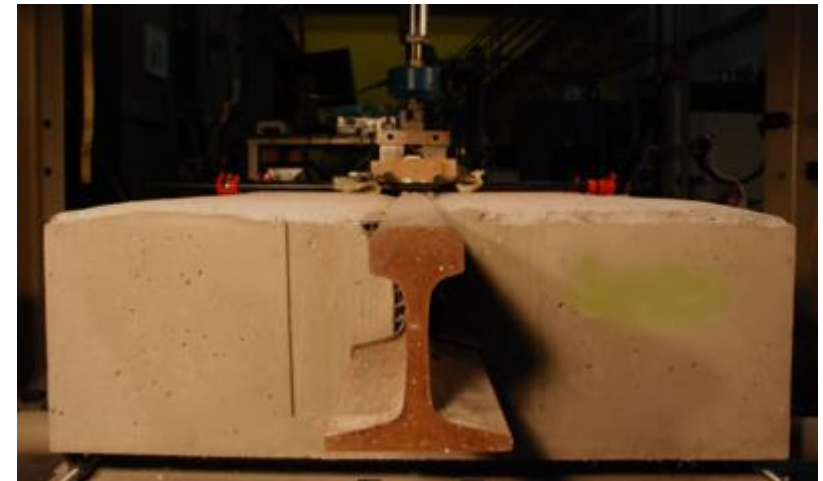
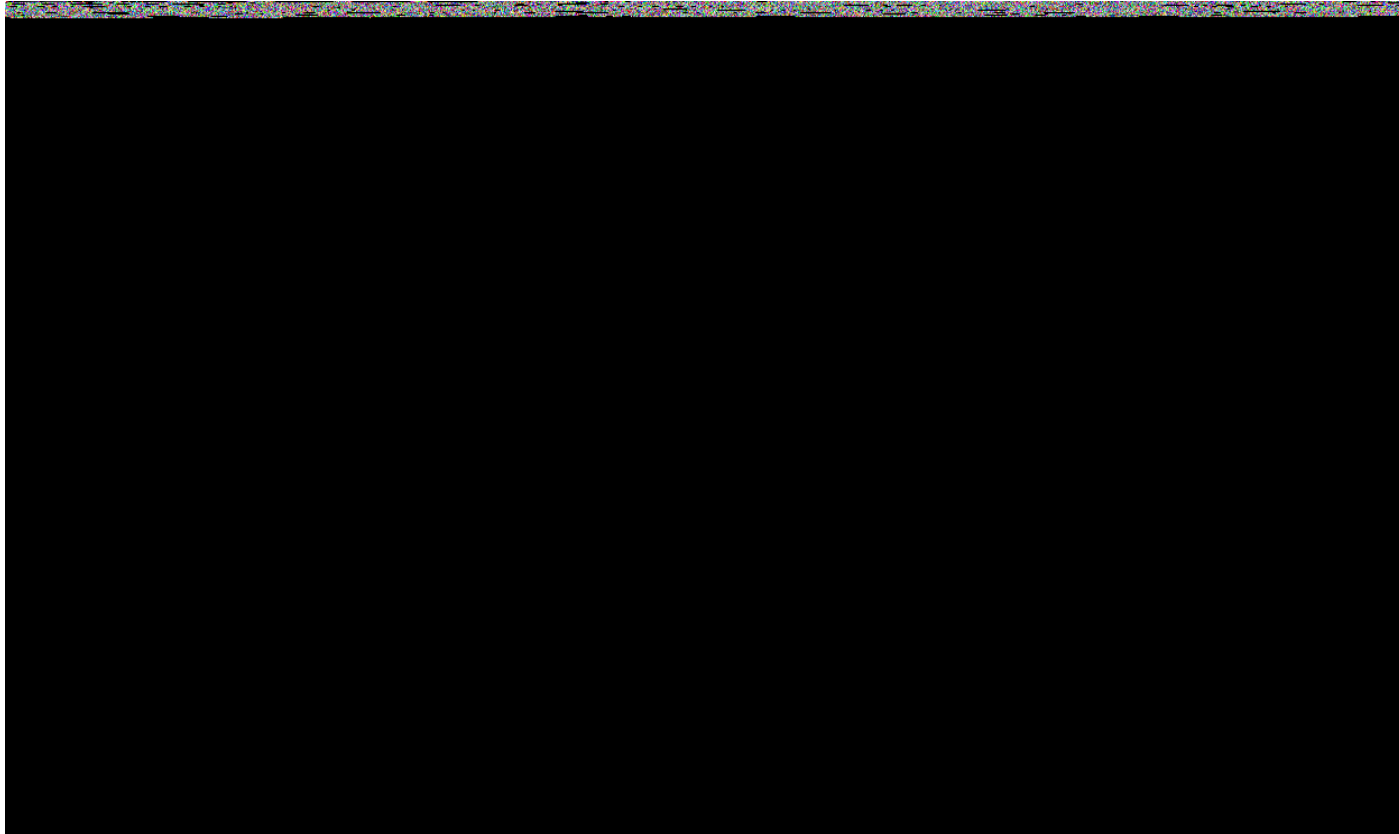
Project Objectives

- Phoenix Metro Light Rail is in the process of expanding the service and adding several stations and track lines.
- The track slabs are 14.5” thick and 8’ wide and made of reinforced concrete sections
- Northwest Phase 2 project with approximately 12,000 feet of the 8’ wide track slab and on South Central project approximately 55,000 feet of the 8’ wide track slab.
- Given the size and cost of the project, the time required for the setting of the rebars, and the volume of the materials used
- It is proposed to consider the cost savings that could be obtained by switching the reinforcement from continuous bars to fibers that are intermixed with the concrete.

Proposed Testing Plan

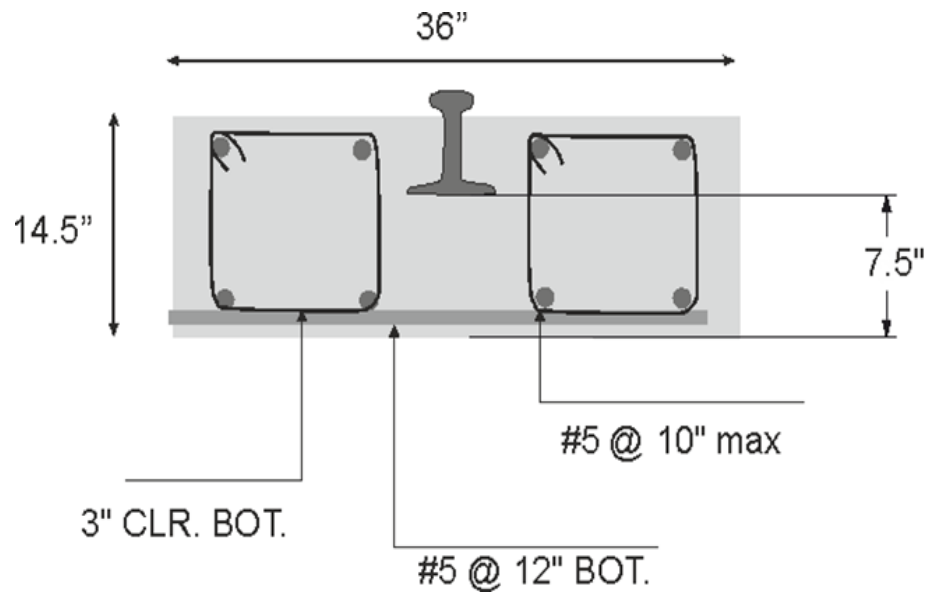
- Develop a Testing Plan for the Mockup design and testing
- Size for the mockups : 3x8 ft by 14.5” for reinforced concrete vs 3x8x 12” with FRC
- Control mockups will contain only the general concrete and the rebar cage used and arranged to meet the same reinforcement ratio of the full-size sections
- Two mockups were made for each of the three test series.
 - Two standard control sections,
 - Two mockups per modified configuration using two different steel fibers supplier designs
- Two suppliers based on their design recommendations were selected. The design alternatives were recommended by the suppliers.

Typical Embedded Track Section

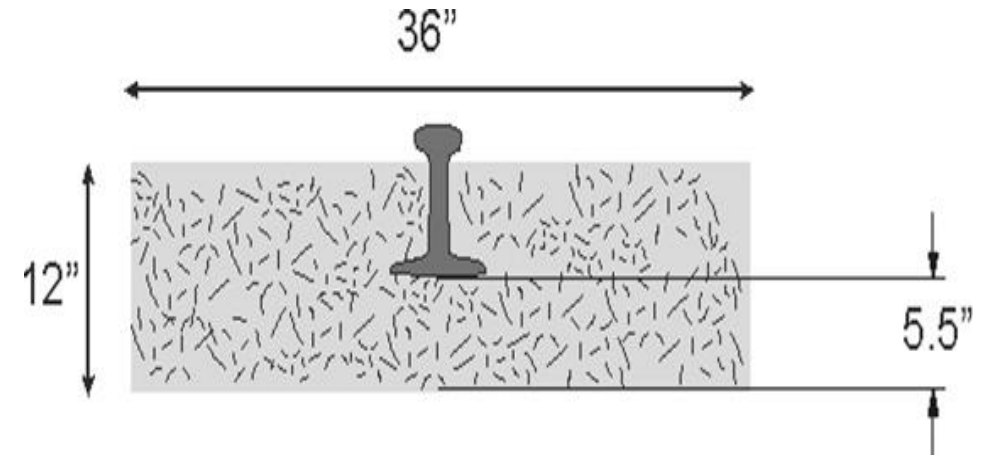


Design Alternatives

Alternative solution to the design of reinforced concrete rail slabs



Conventional Design section



FRC Section

Project was in collaboration with:
Kiewit McCarthy, a Joint Venture (NWE2)
3888 East Broadway Road, Phoenix, AZ 85040
Mr. Gary Sanders, Project Manager

Mock-ups representing Current vs. Proposed Design



Construction of current Design approach consumes significant time



Proposed design expedites the construction time and cost significantly.

Attachment of Strain Gages



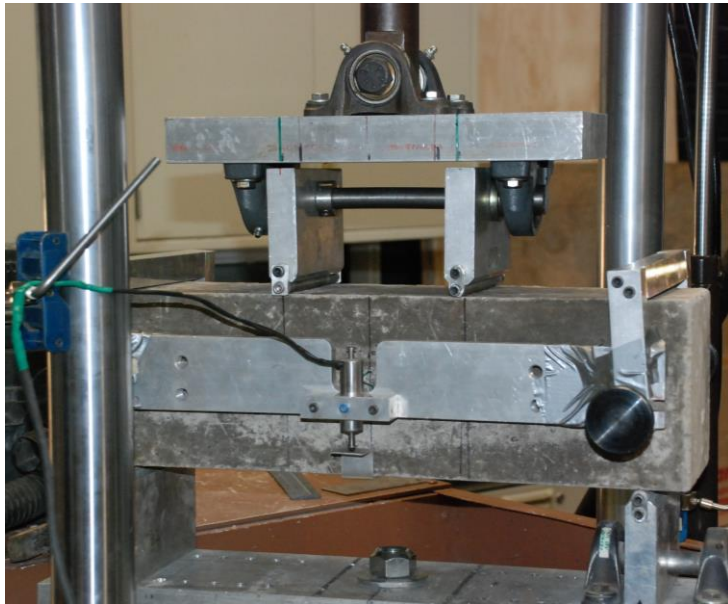
Sample Preparation



Objective of fullscale testing: Validate use of FRC

Valley Metro, Light Rail (PHX)

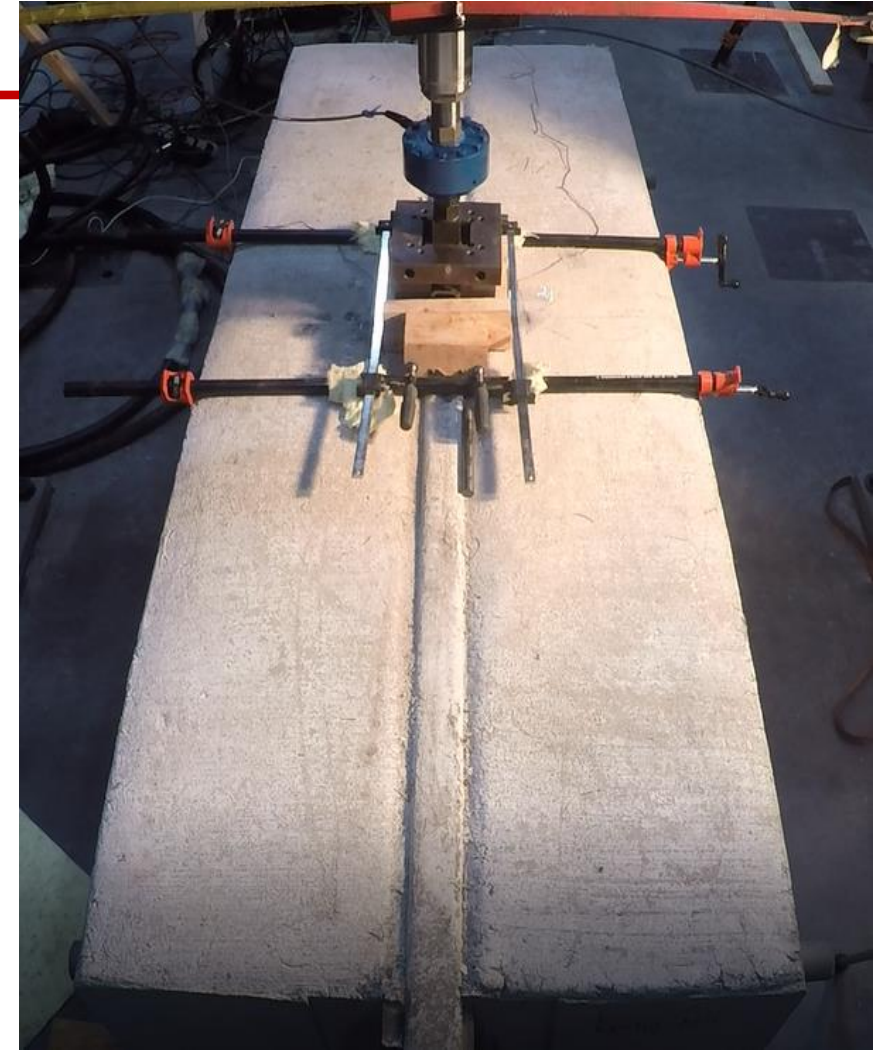
1. ASTM C1609 x 6
2. Monotonic x 3
3. Fatigue x 4



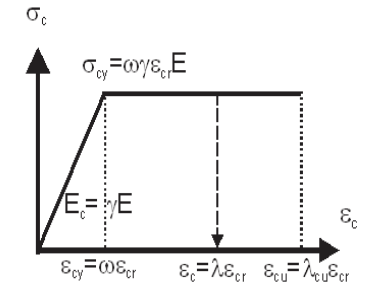
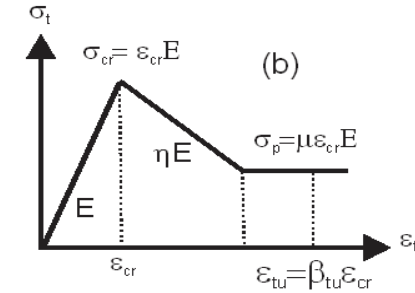
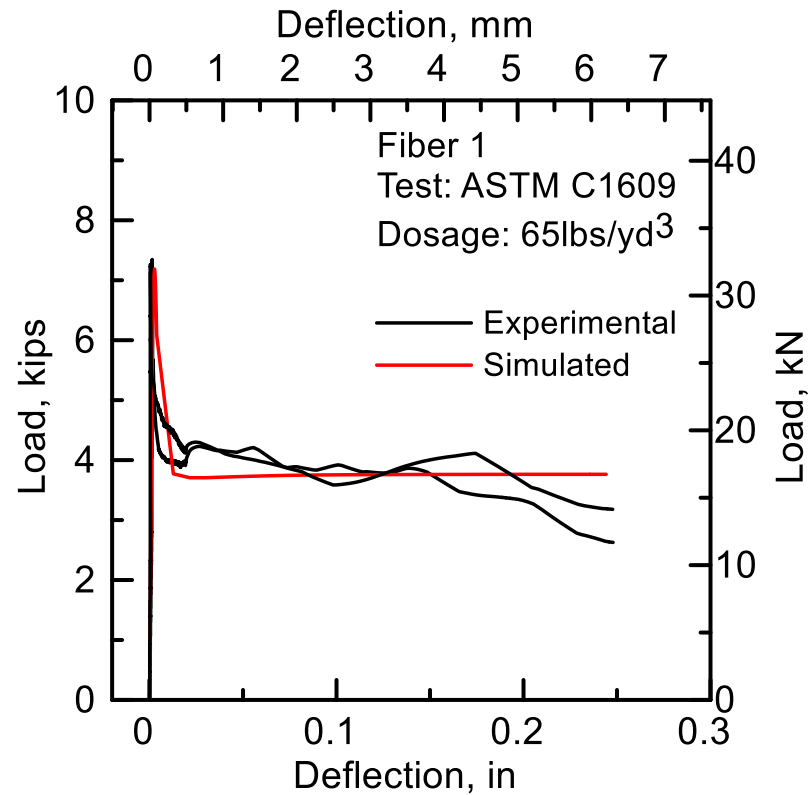
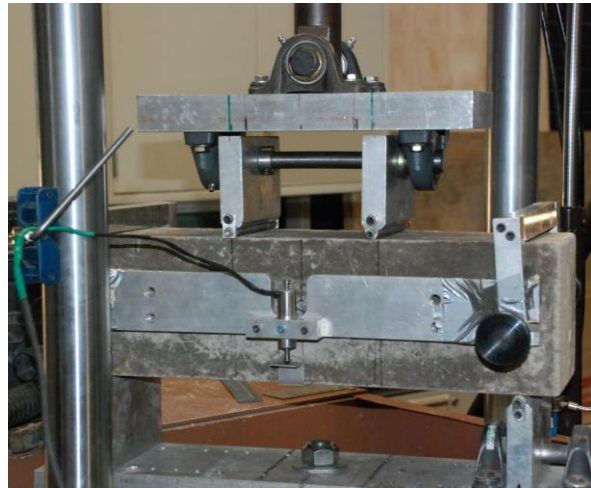
ASTM C1609



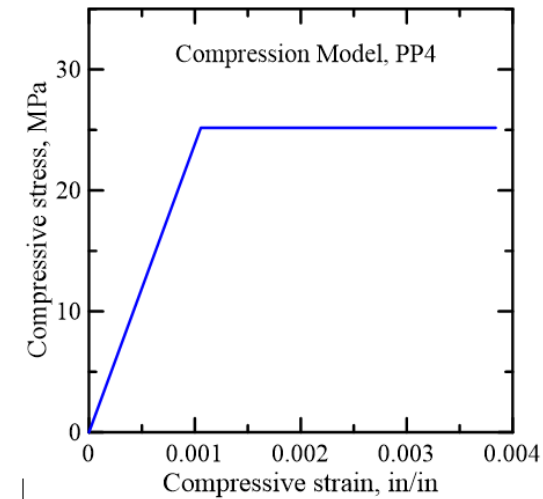
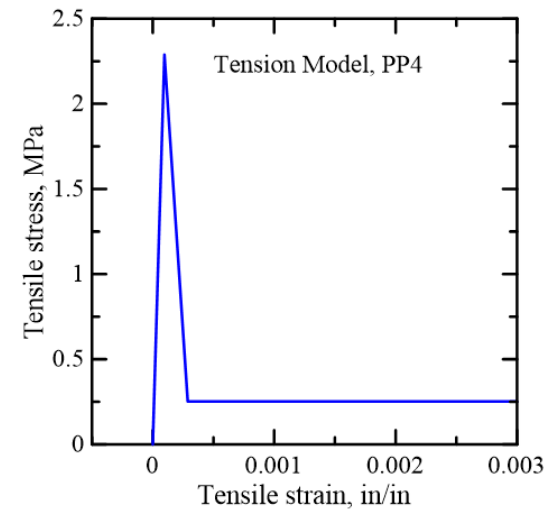
ASTM C39



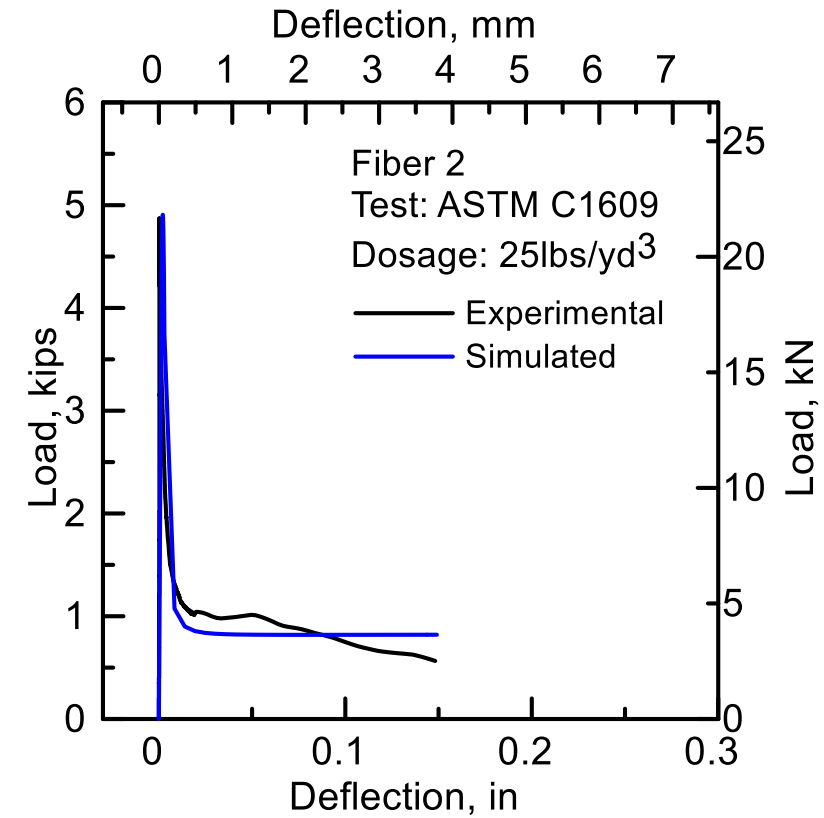
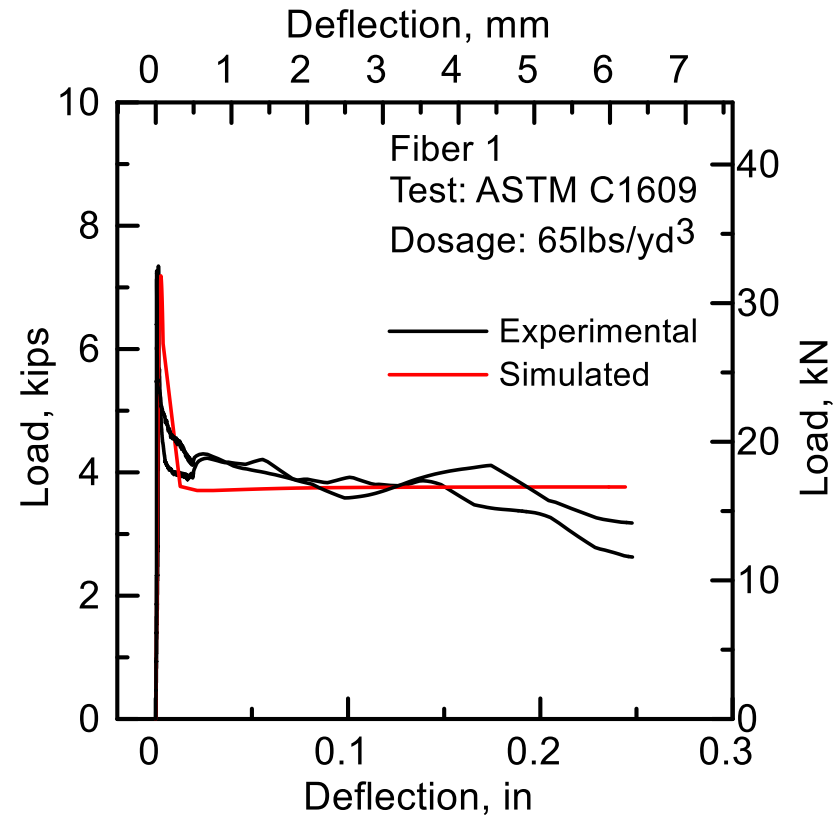
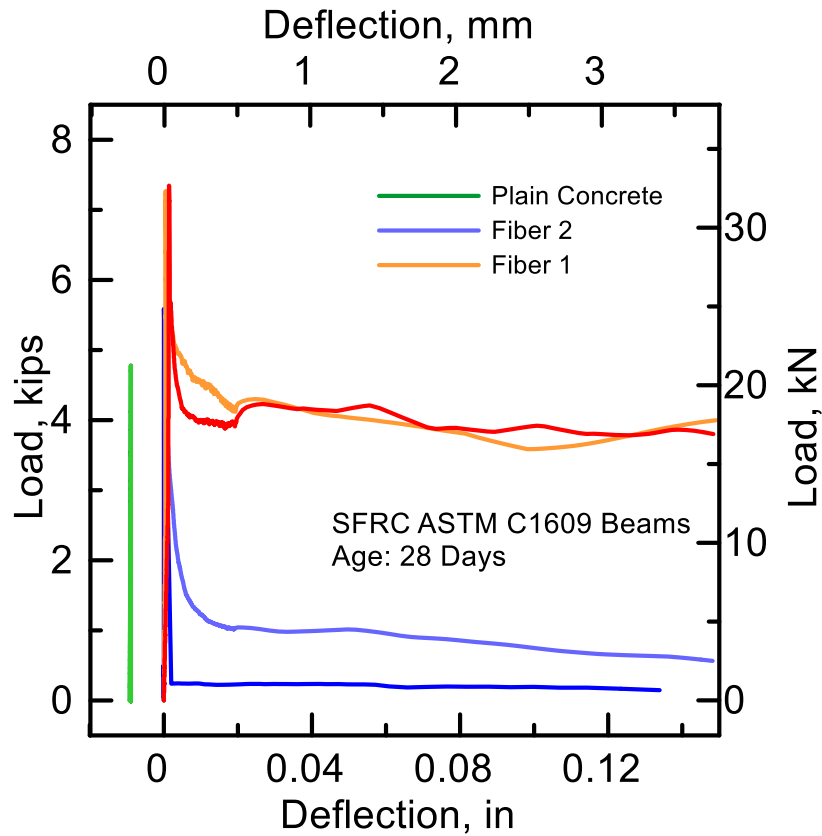
Standard Design Procedure using ACI 544.4R method (Parametric-based design)



Using these parameters, the location of neutral axis, moment capacity, and curvature of the section can be obtained at any imposed strain



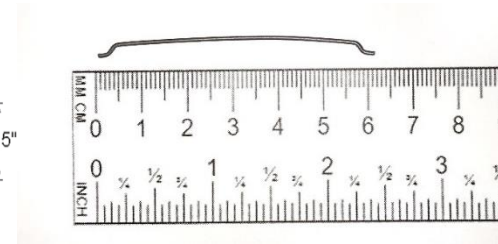
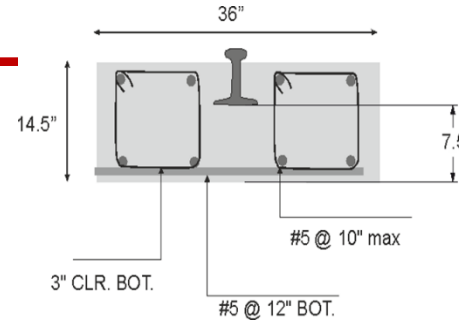
Material Property tests



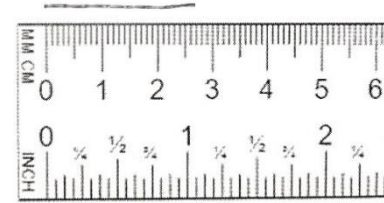
Full scale mockup specimen testing

3 type of mockup sections

- Conventional Rebar section (14.5"x3'x8')
- FRC, Steel Fiber, #1, 65 PCY Dosage (12"x3'x8')
- FRC, Steel Fiber #2, 25 PCY Dosage(12"x3'x8')



Type 1



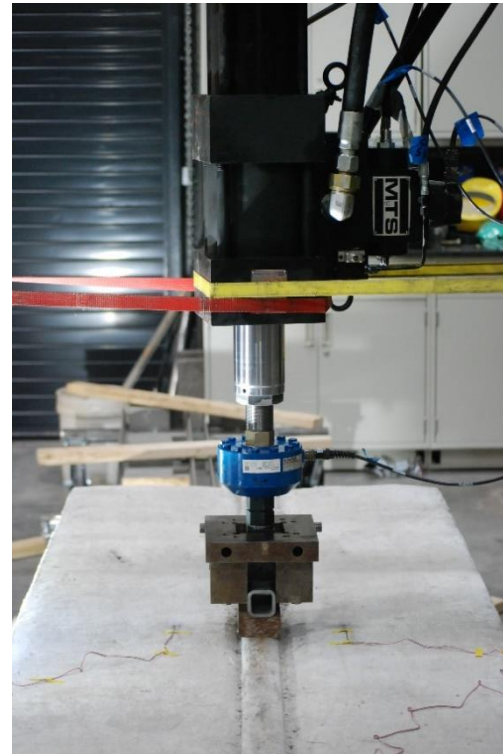
Type 2

Testing Plan

- Monotonic (3PB flexure) testing of mockup specimens
- Fatigue testing at 4Hz loading frequency

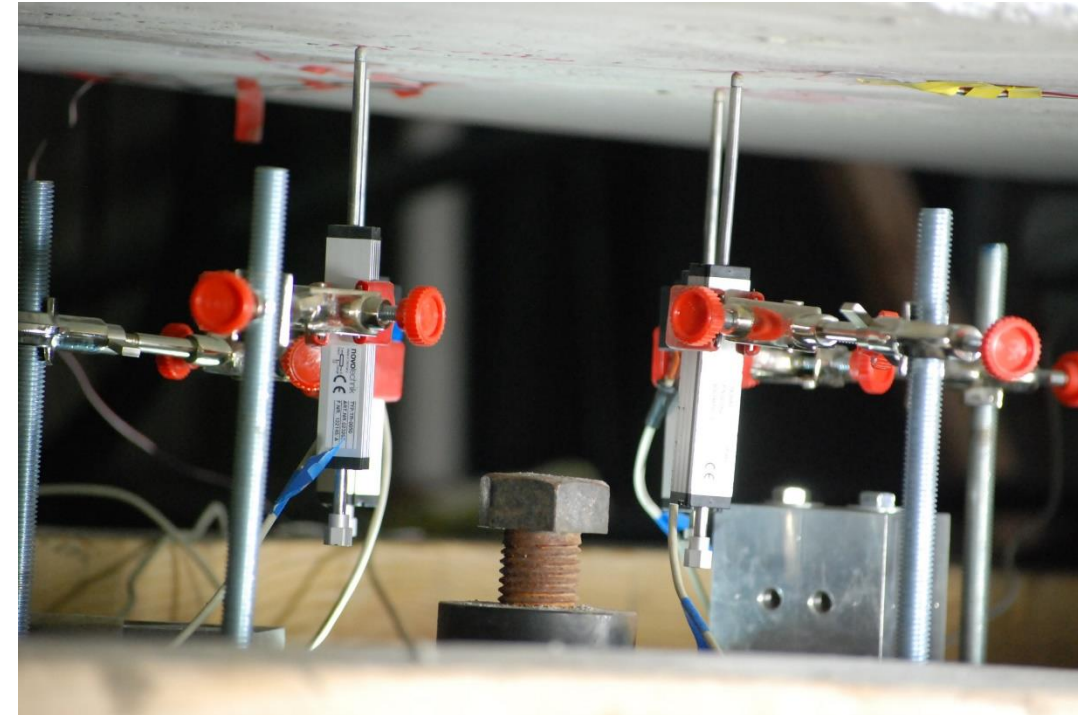
Instrumentation

- 4 LVDT at bottom
- 4 strain Gages (2 @ Compression zone, 2 @ Tension zone)



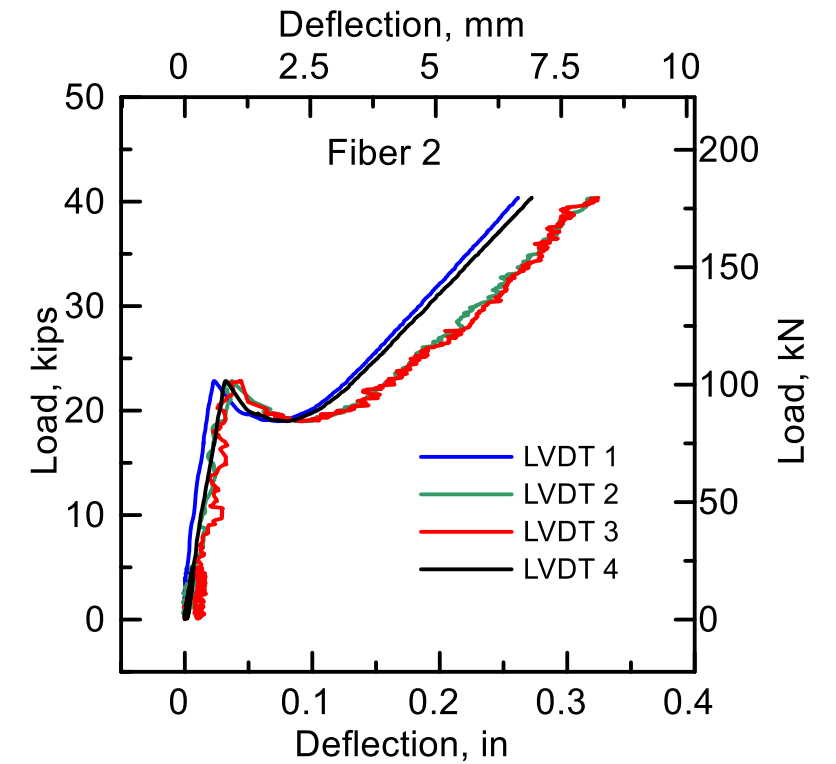
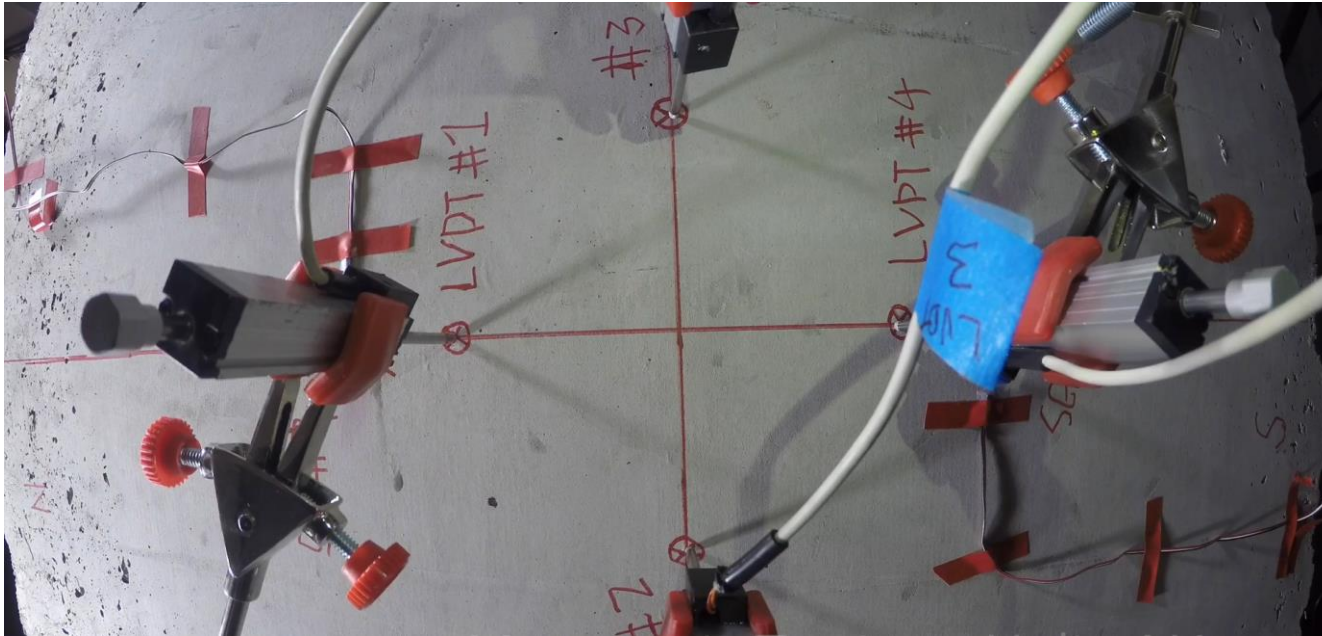
55 kip load frame setup

Full size Slab testing

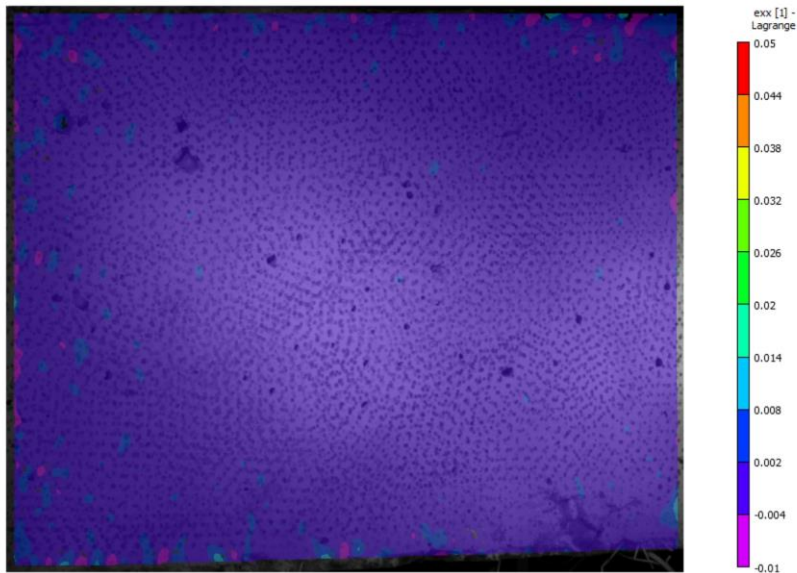


Underneath the sample showing the deflection measuring instrumentation

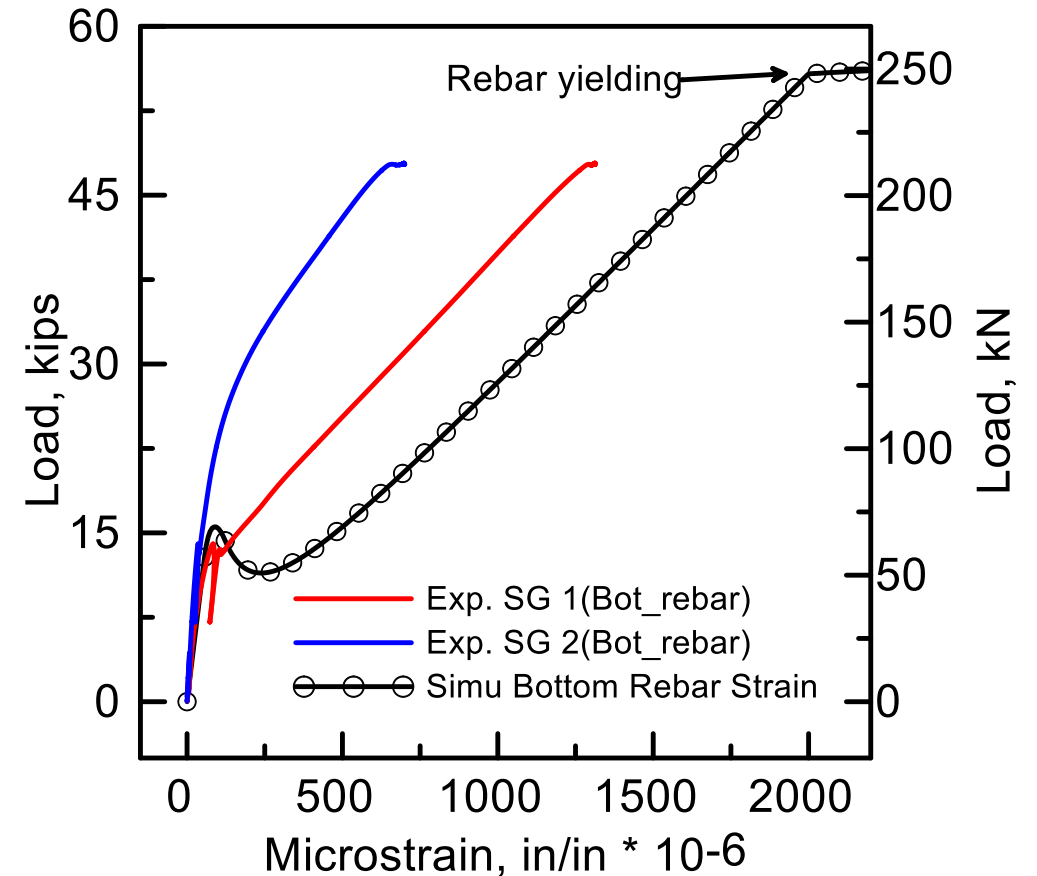
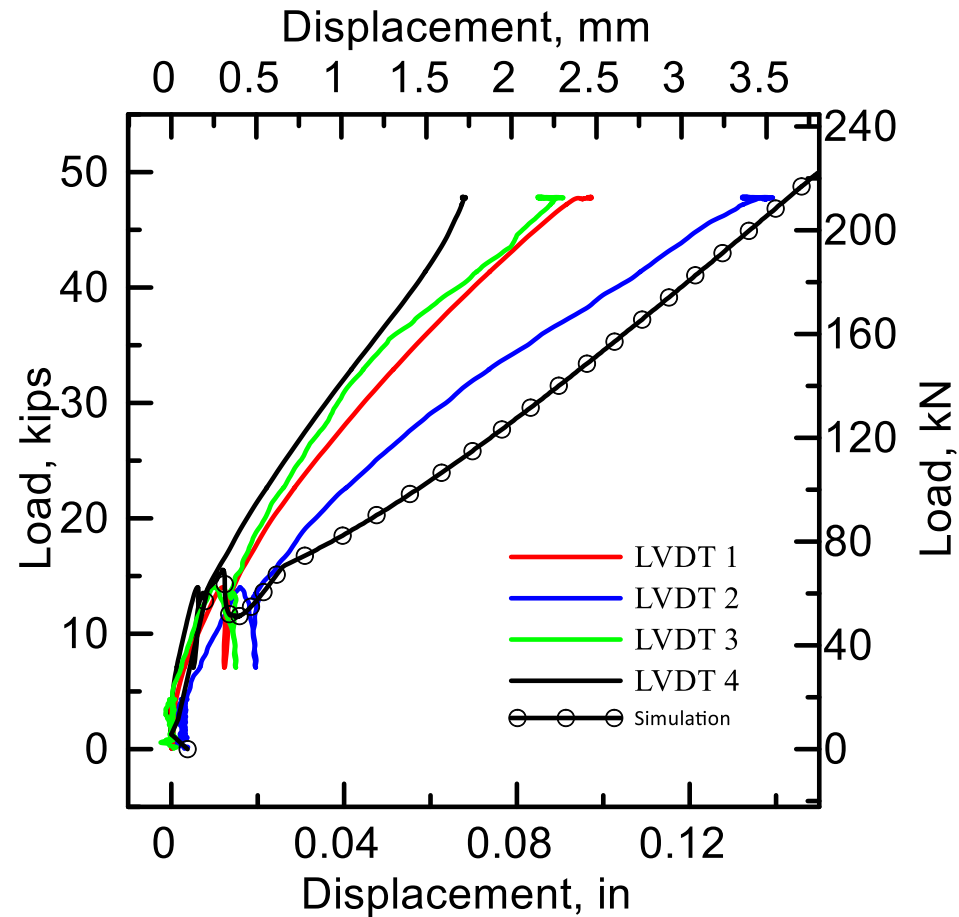
Full scale mockup specimen monotonic testing



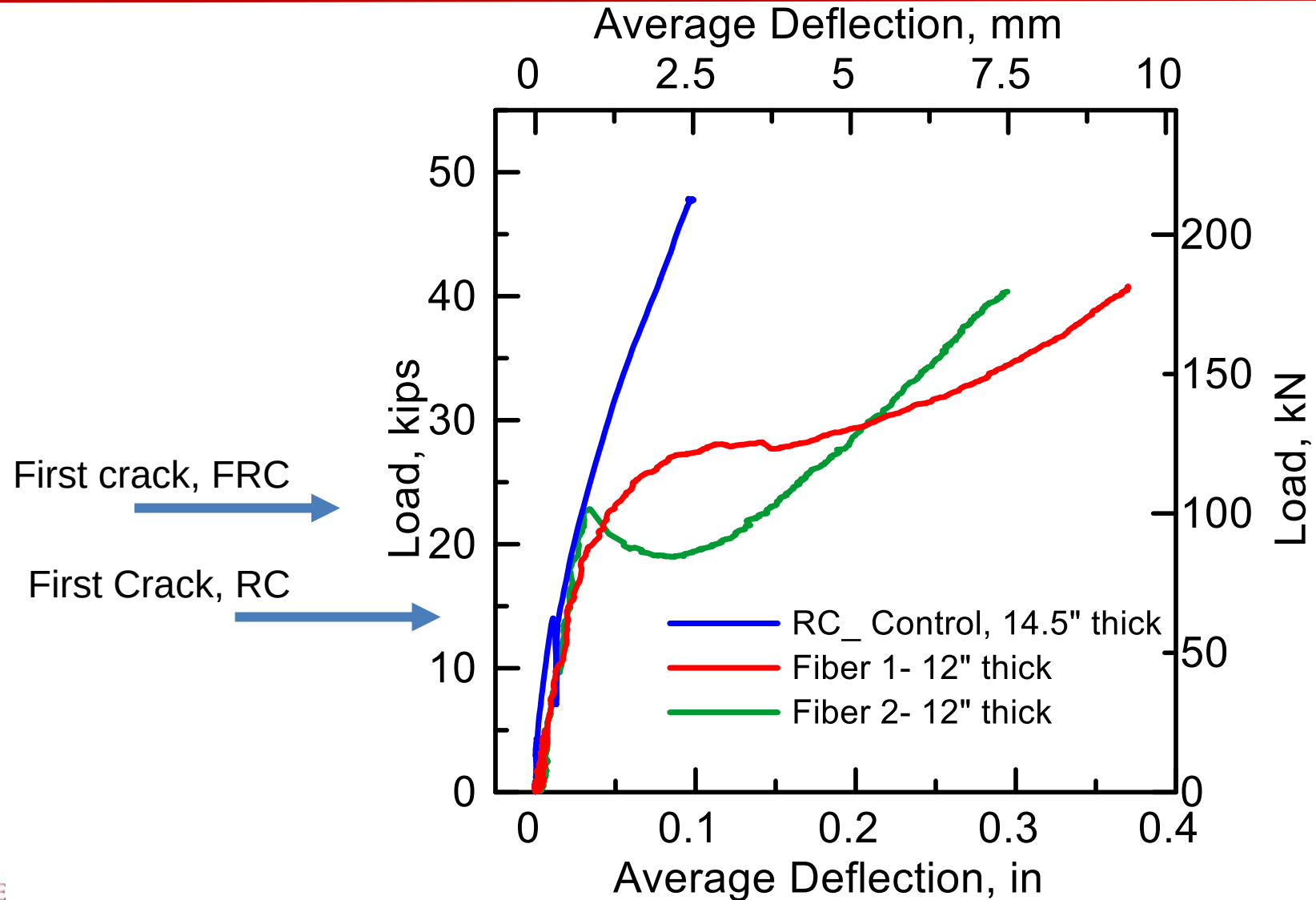
DIC technique to document crack opening



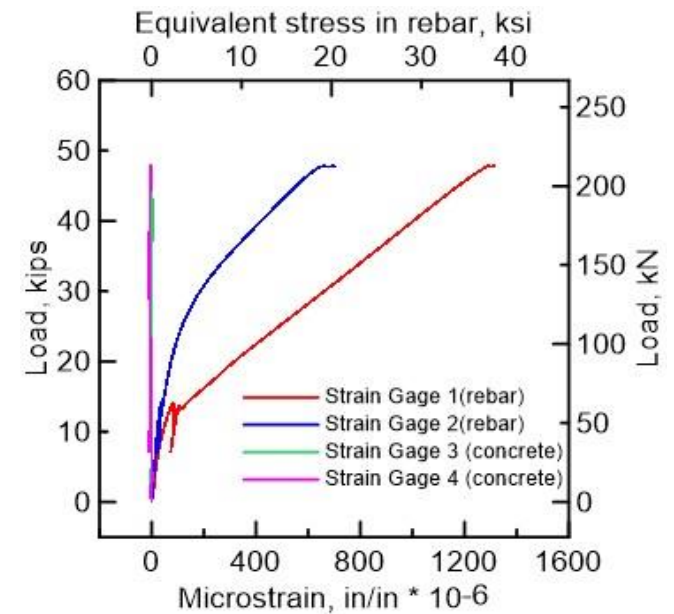
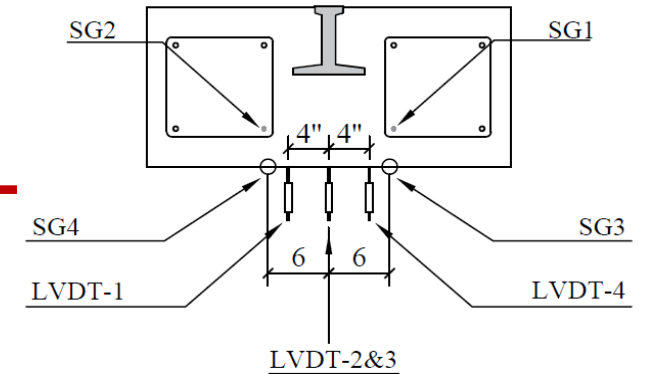
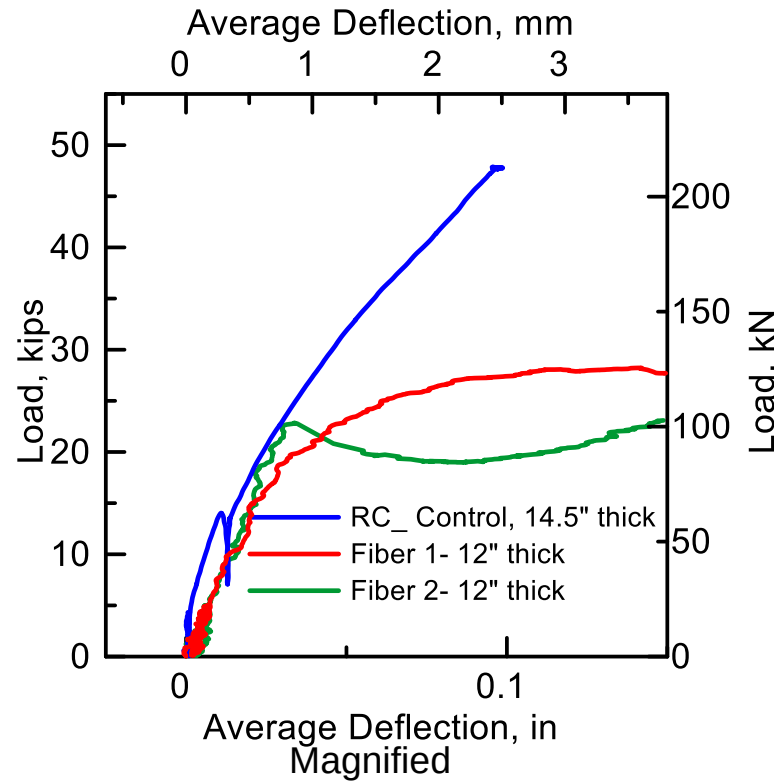
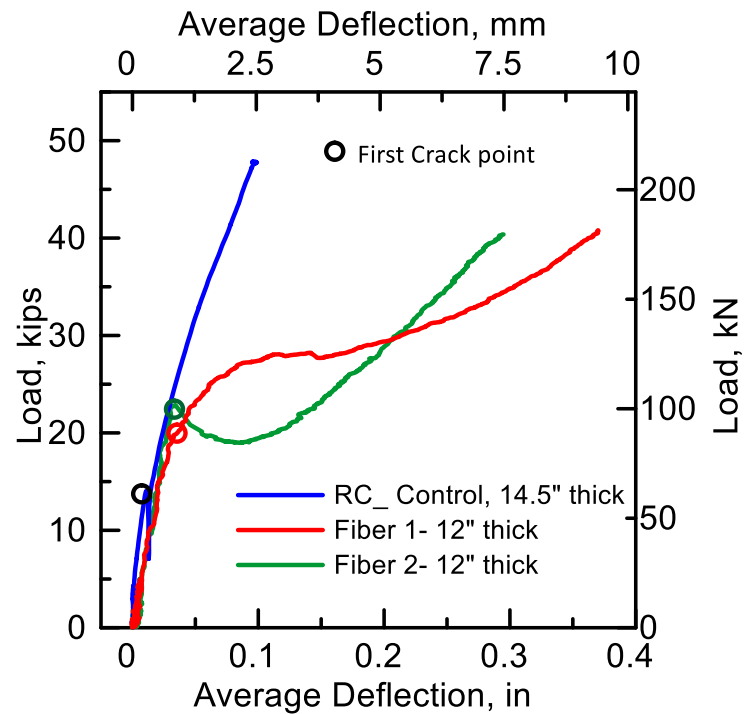
Standard Control Sample



Comparison of three alternatives



Summary of Full scale mockup specimen testing

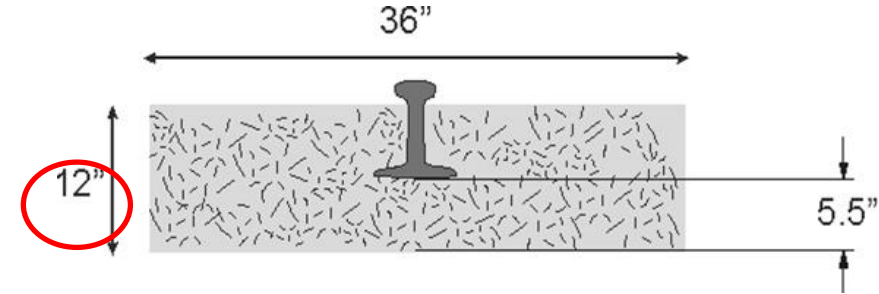
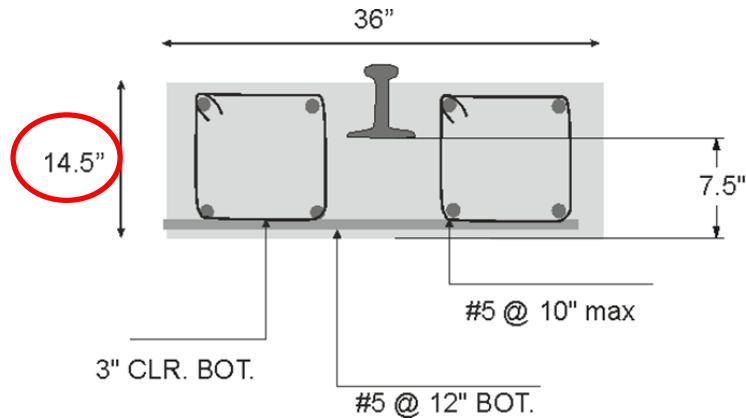


RC1 Specimen

Observations

- Rebars still in elastic phase at 50 kips loading, 225 kN
- Higher post crack load carrying capacity in Fiber 1 Specimen (65pcy) Compared to Fiber 2 (25 pcy)

Monotonic Test Results



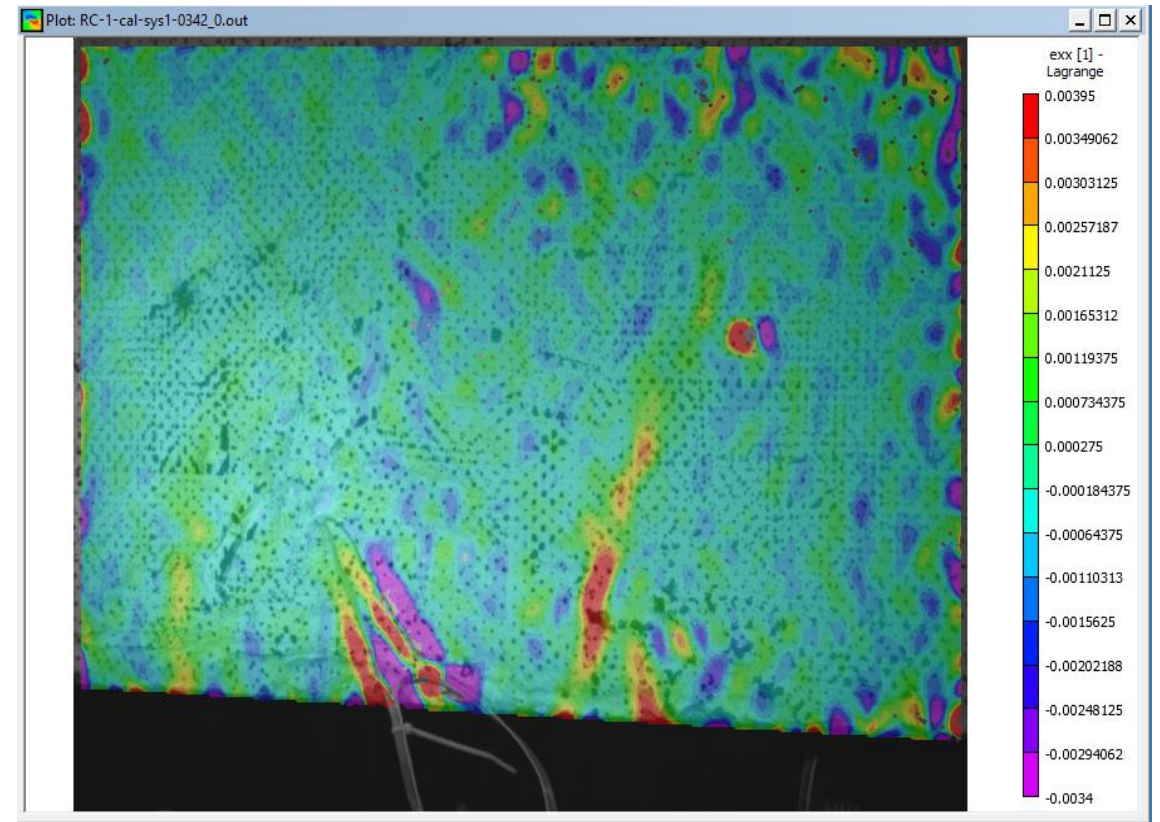
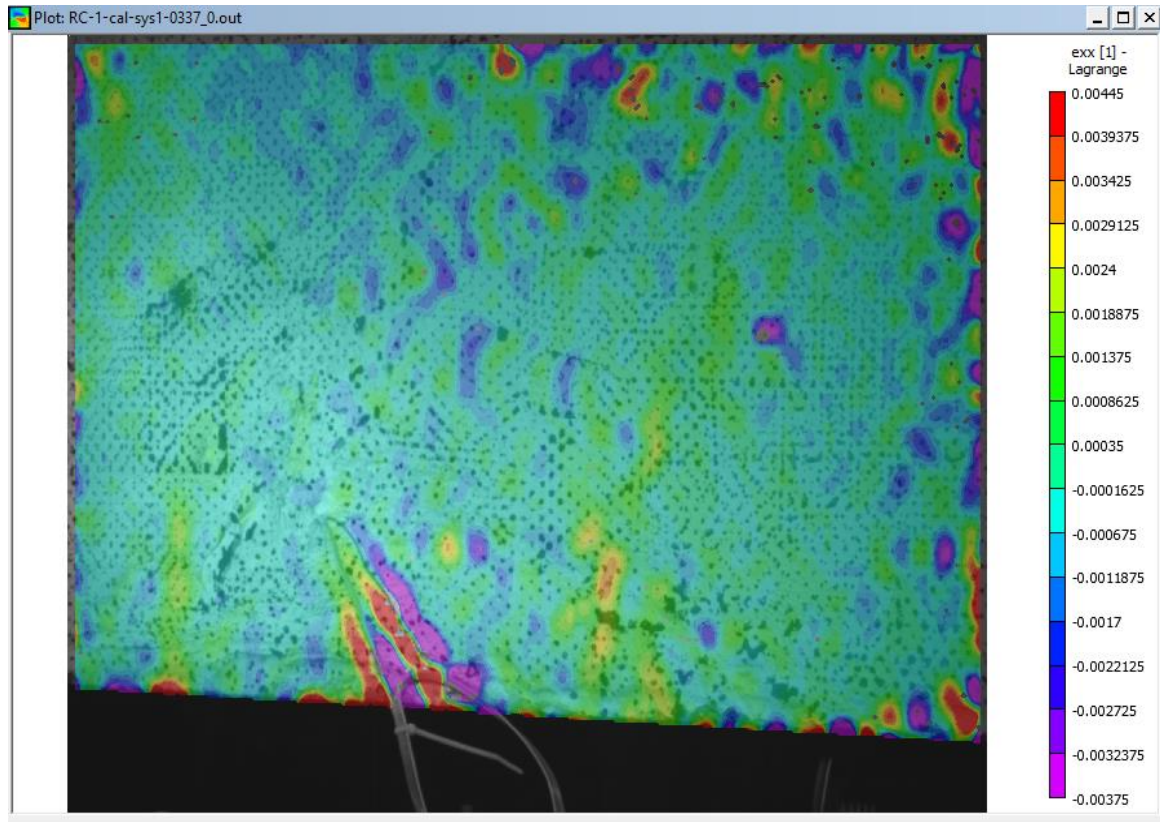
The relative stiffness of RC and FRC = $\frac{I_{RC}}{I_{FRC}} = \frac{b \times (14.5)^3 / 12}{b \times (12)^3 / 12} = \frac{14.5^3}{12^3} = 1.76$

The 75% increase is compatible with the almost twice stiffness of the RC specimen compared to the FRC specimen.

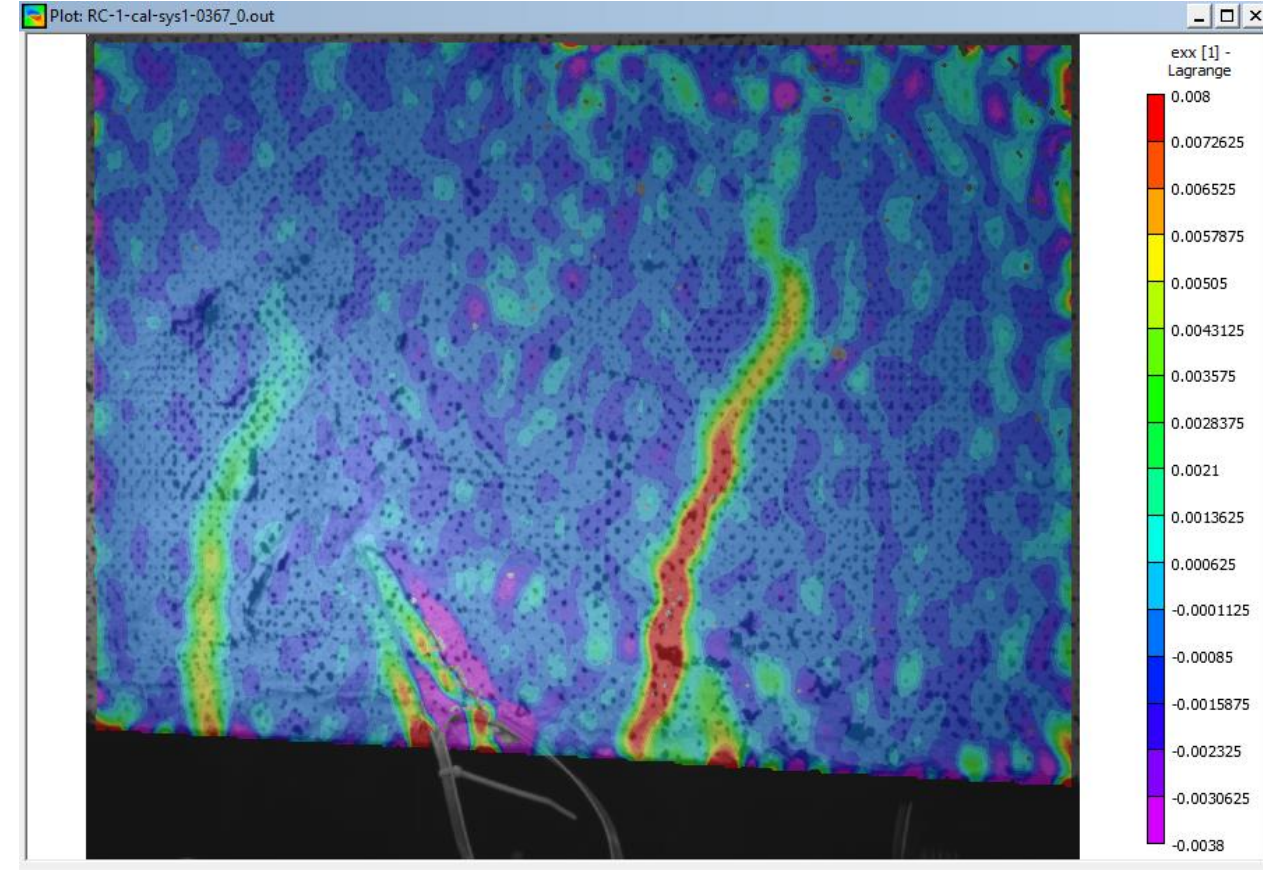
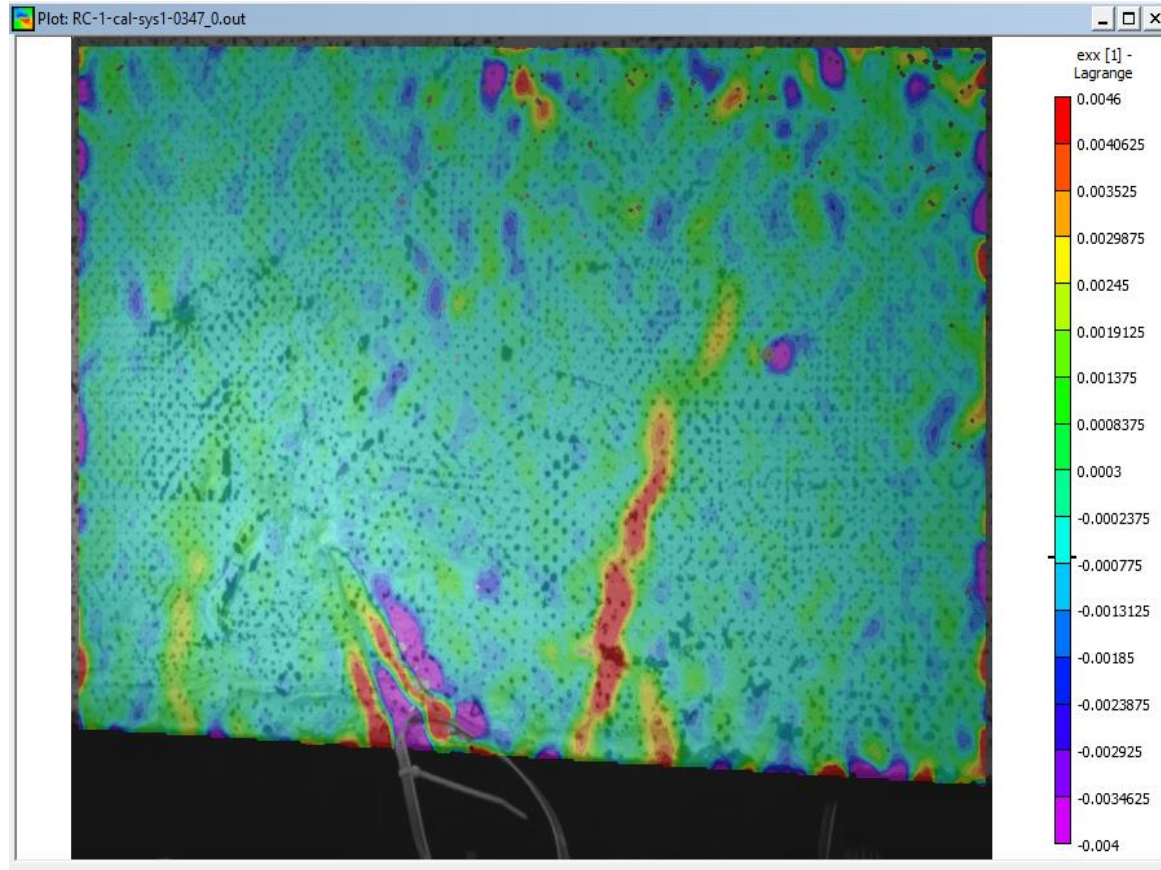
Observations

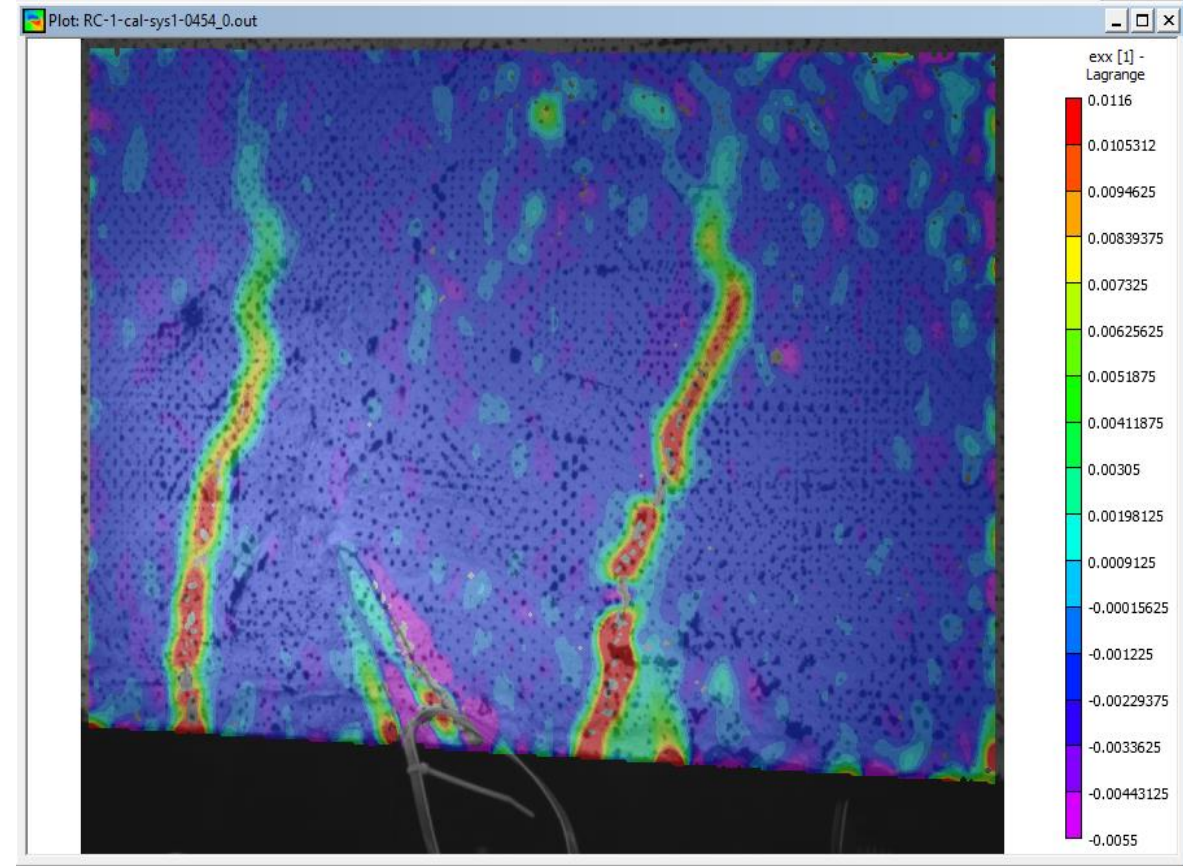
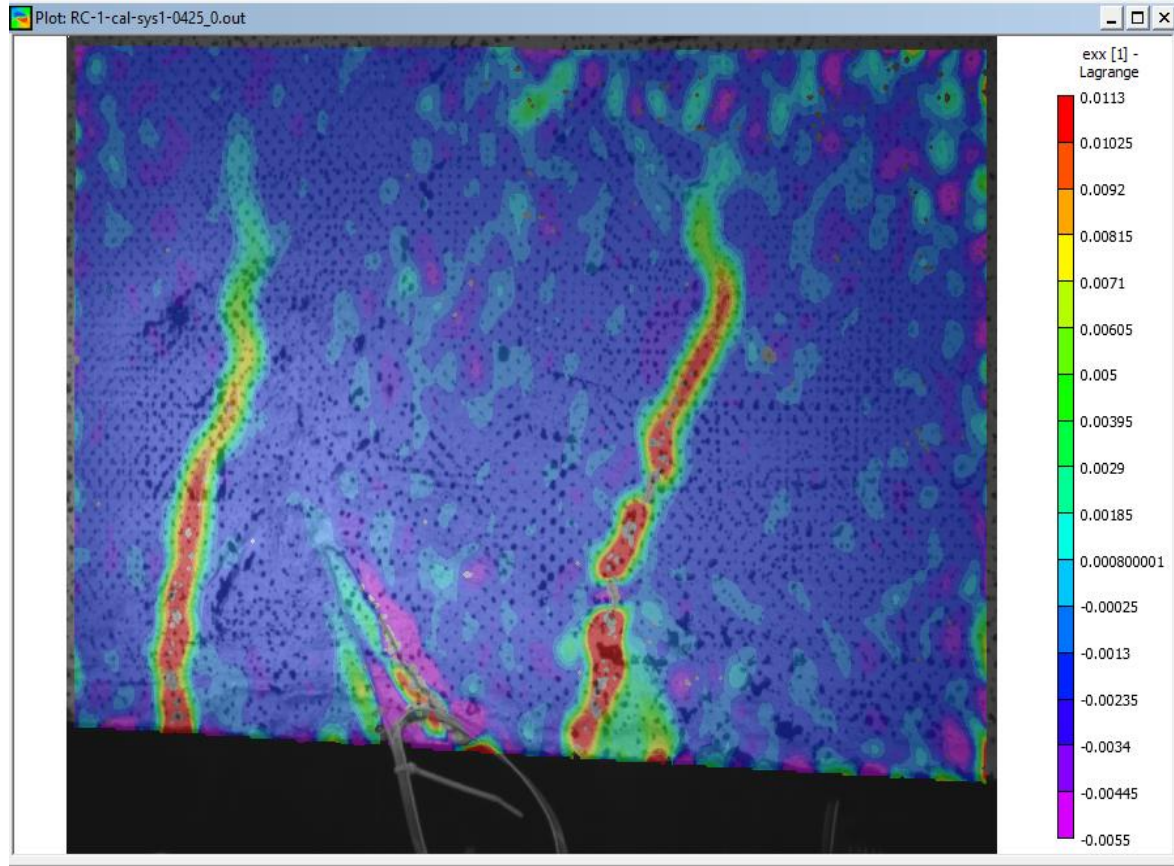
- In RC specimen rebars still in elastic phase at 50 kip loading
- Higher post crack load carrying capacity in Fiber 1 Specimen (65PCY) Compared to Fiber 2

DIC Data collection

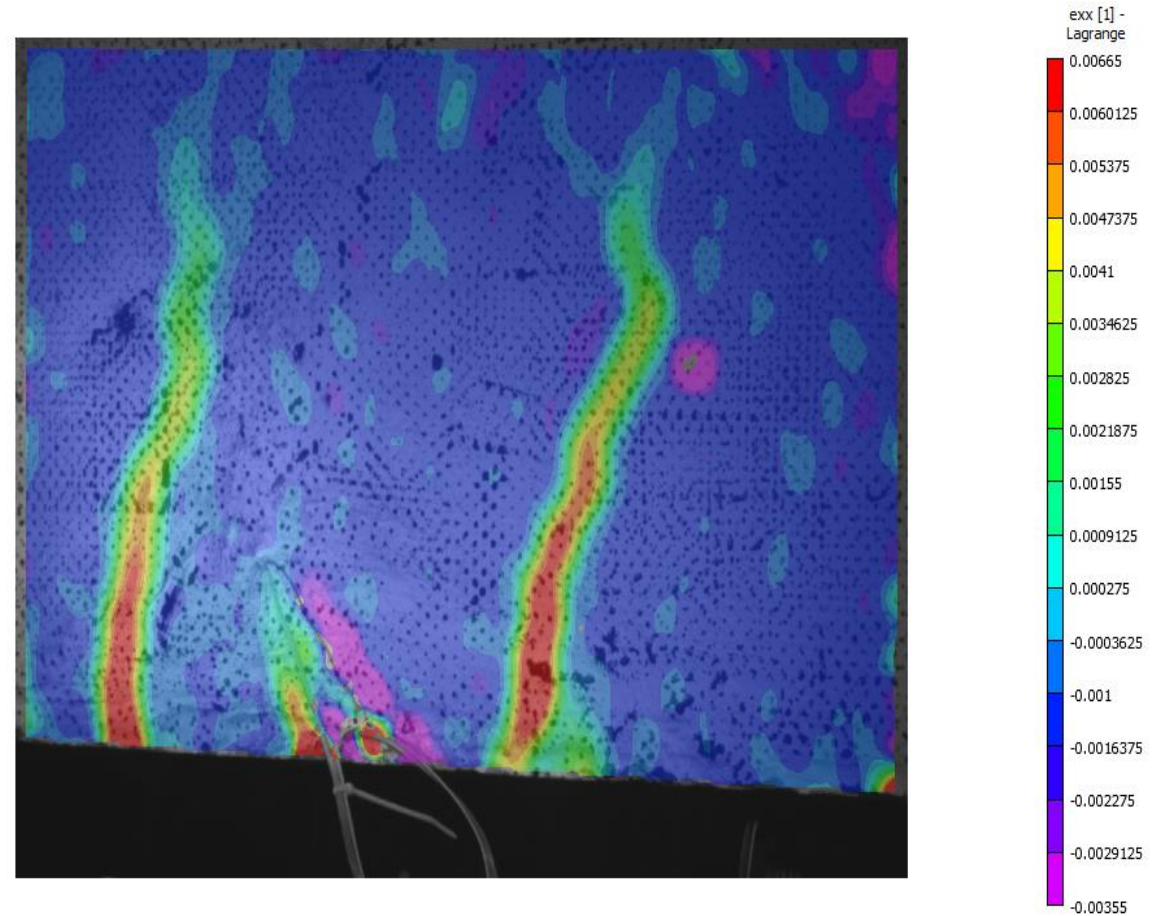
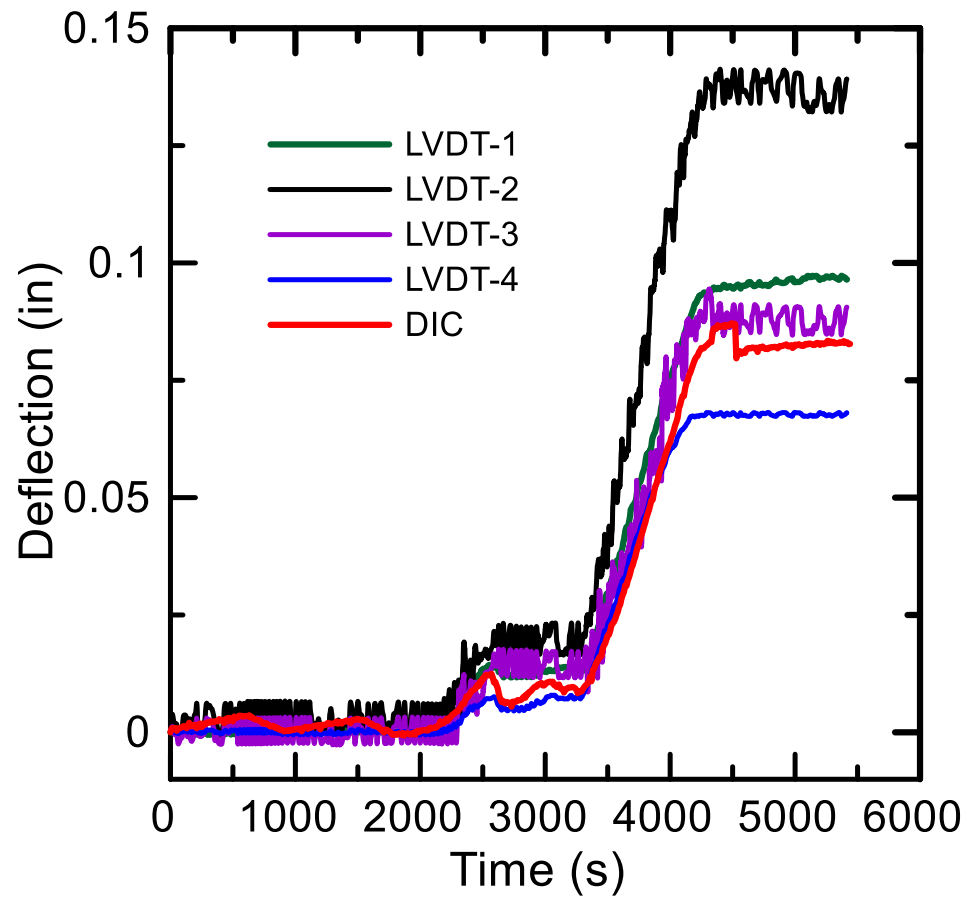


Various stages of cracking

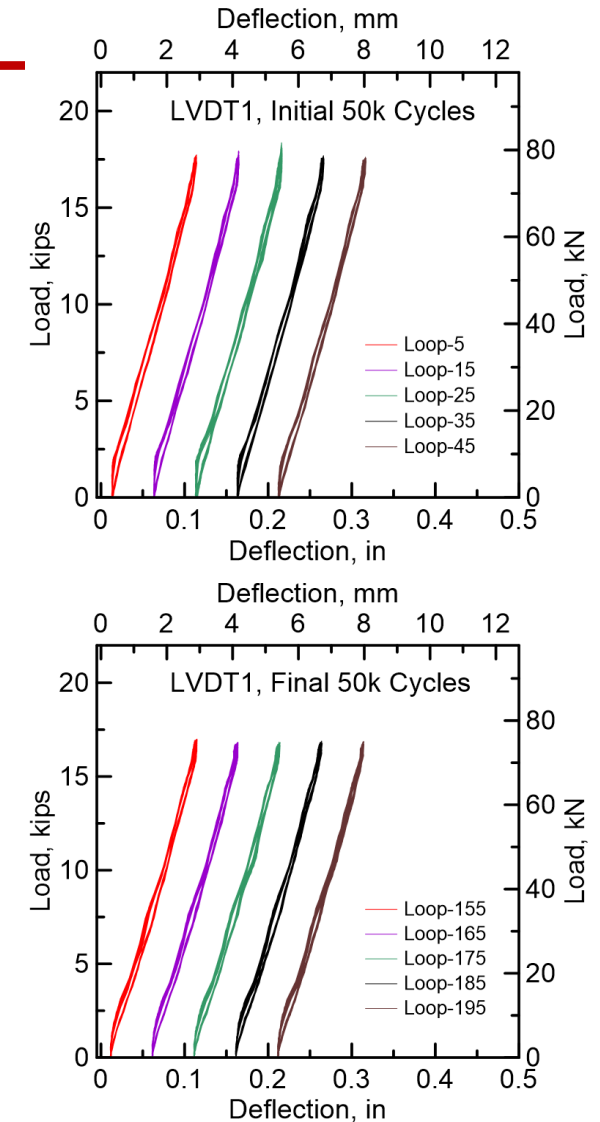
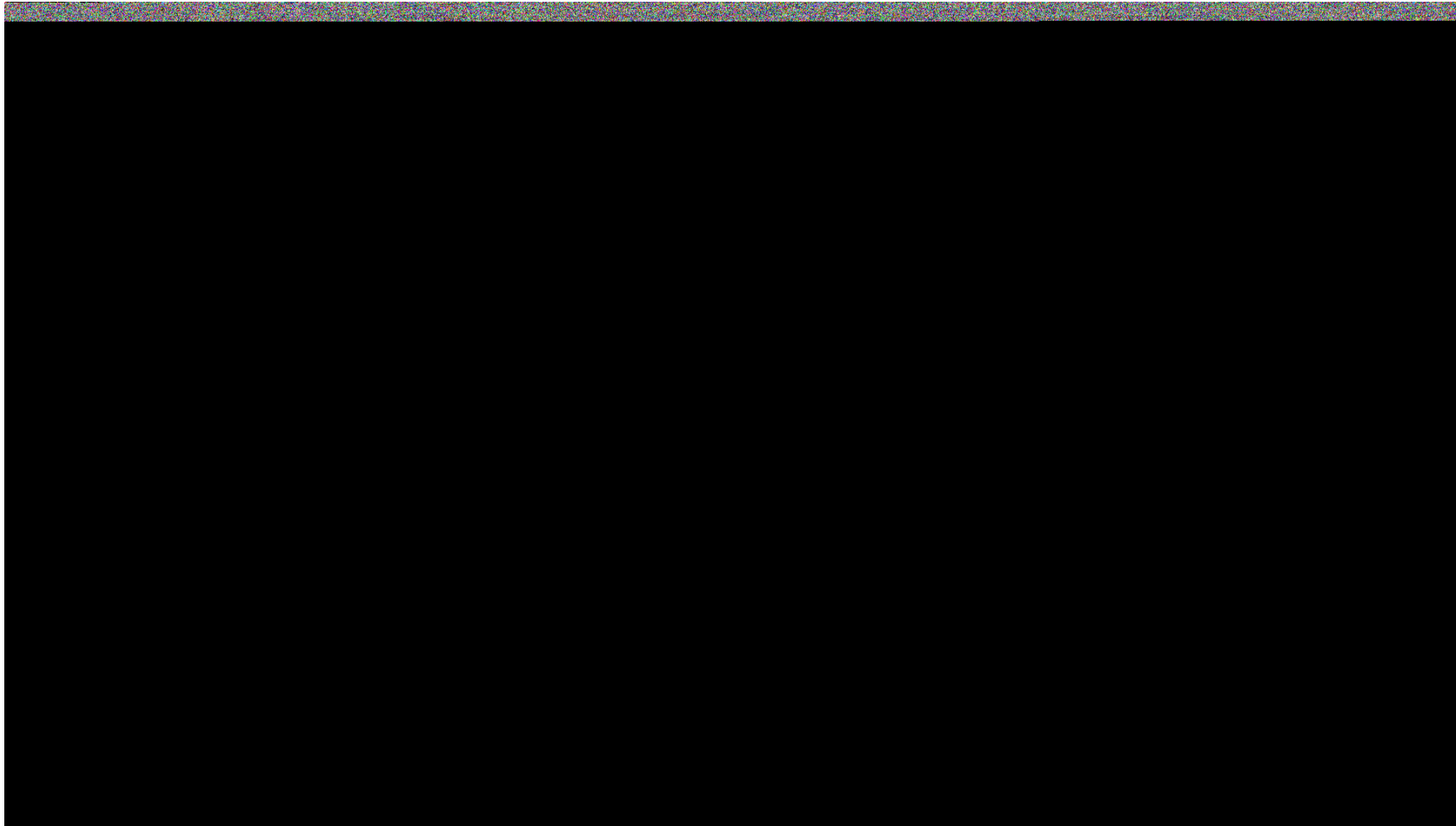




Verification of displacement with DIC and the final state of cracking



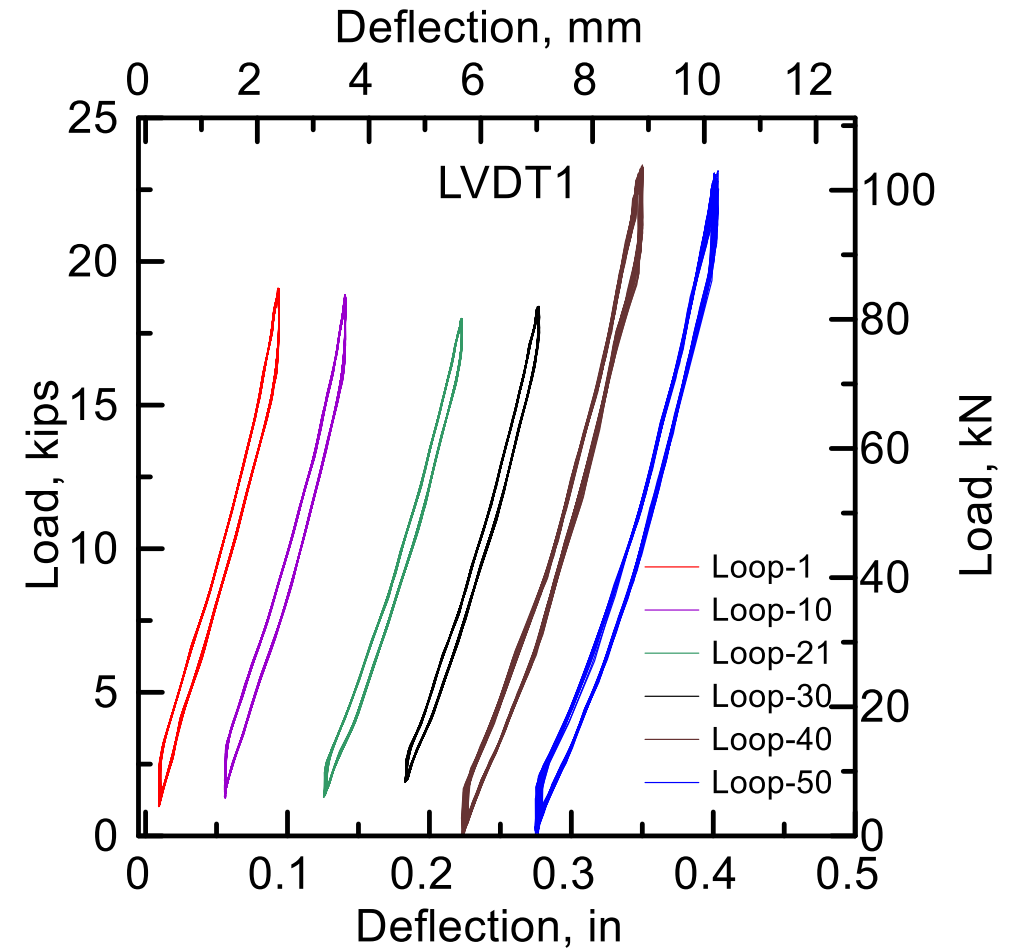
Fatigue Testing (2 Million Cycles , 4Hz)



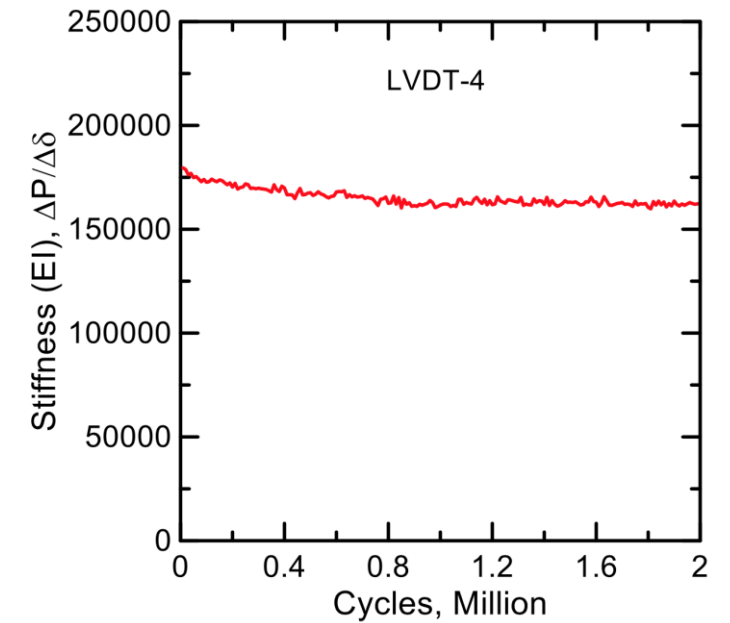
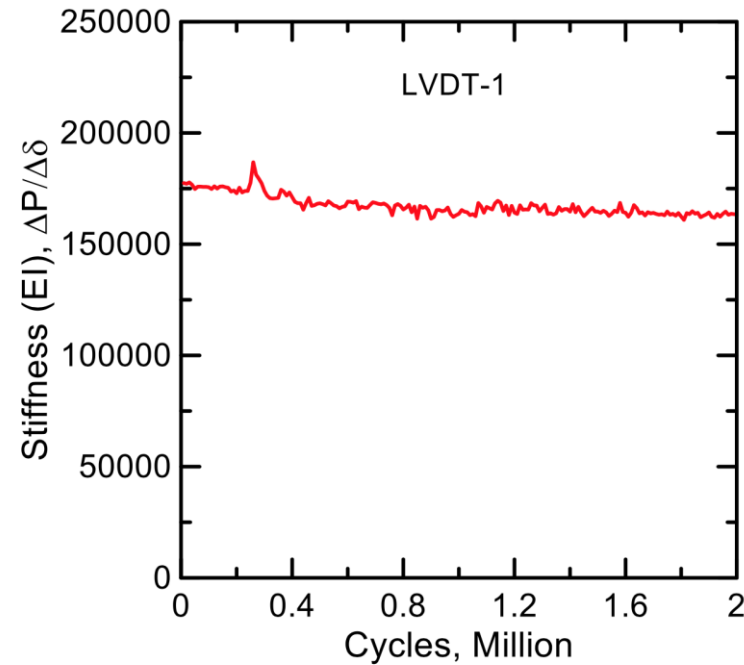
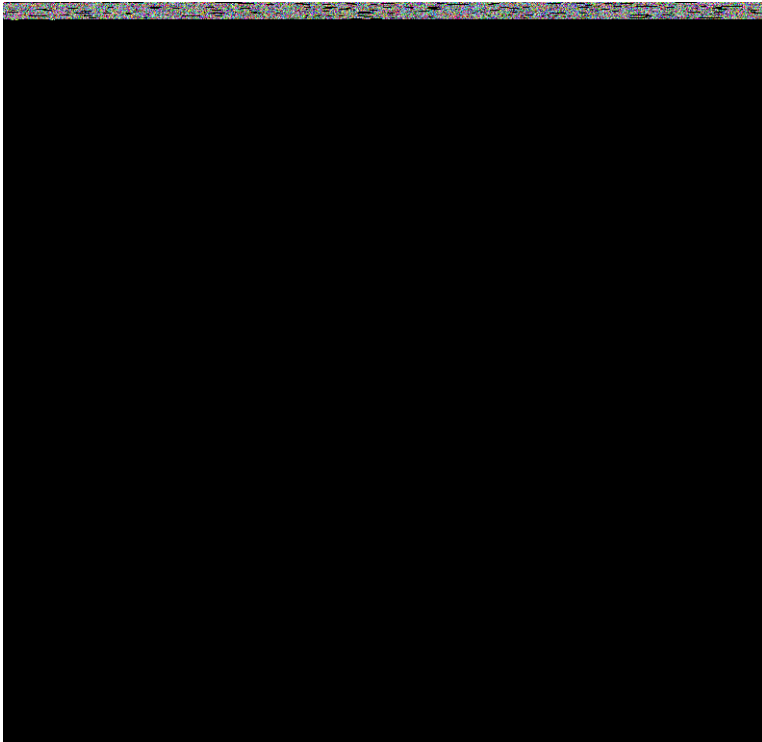
- Loading : 17.5 kips , 80 kN
- Data Acquisition of 100 cycles per 10,000 cycles, 2.0 million cycles, 8 days at 4 Hz

Fiber 1,

- RC sample test terminated after 200k cycles
- Fiber 2 terminated after 150k Cycles
- Fineber one completed after 2M Cycles

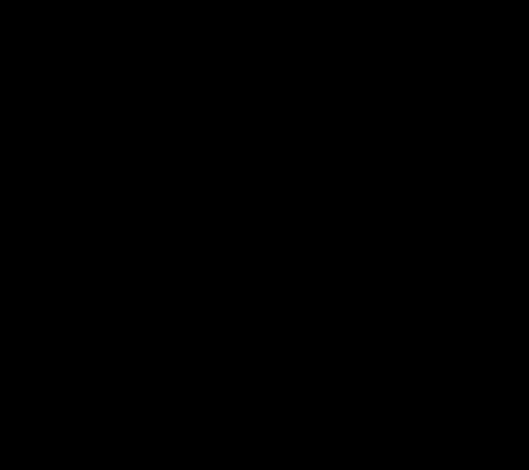


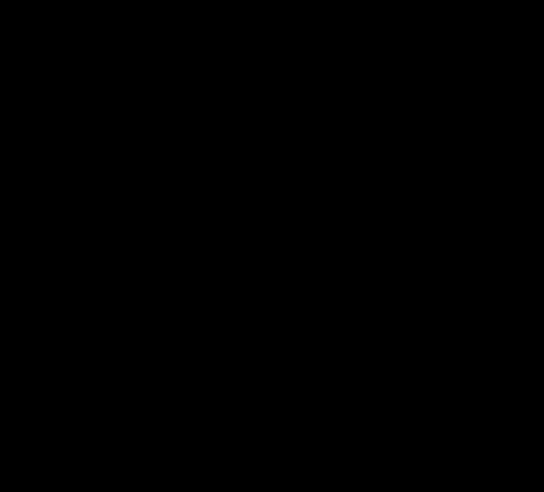
Full scale Fatigue Test Results

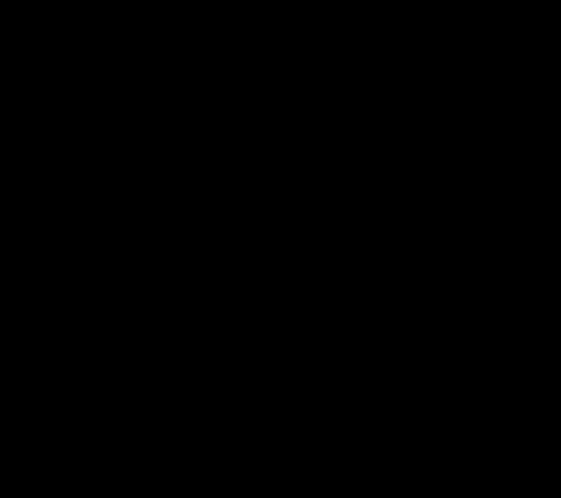


Observation

-Average decrease in stiffness of 6.83% after 2 million cycles

$$P(\delta) = P_s(\delta) + P_c(\delta)$$


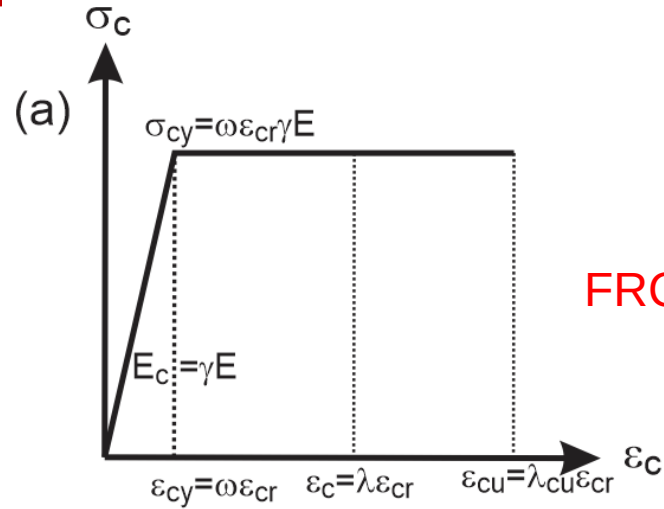




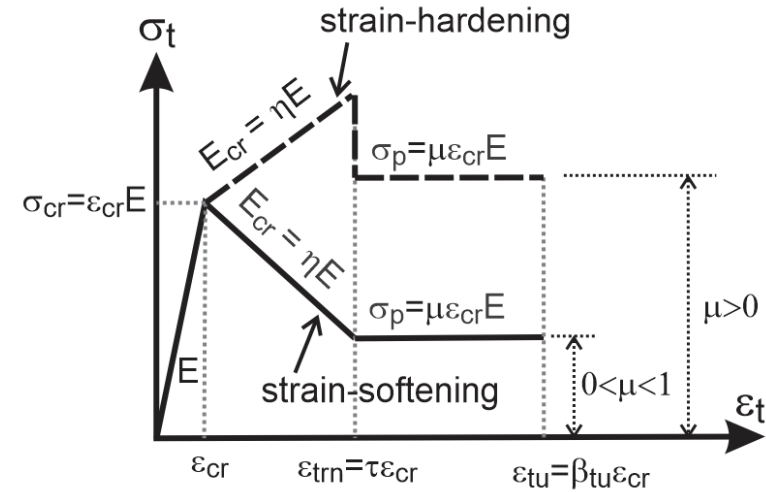
55



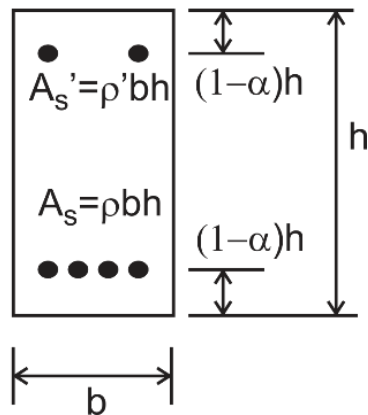
ASU Constitutive model



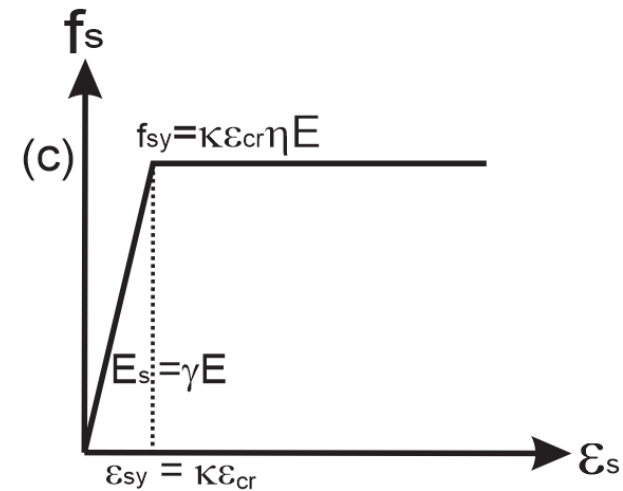
FRC Compression



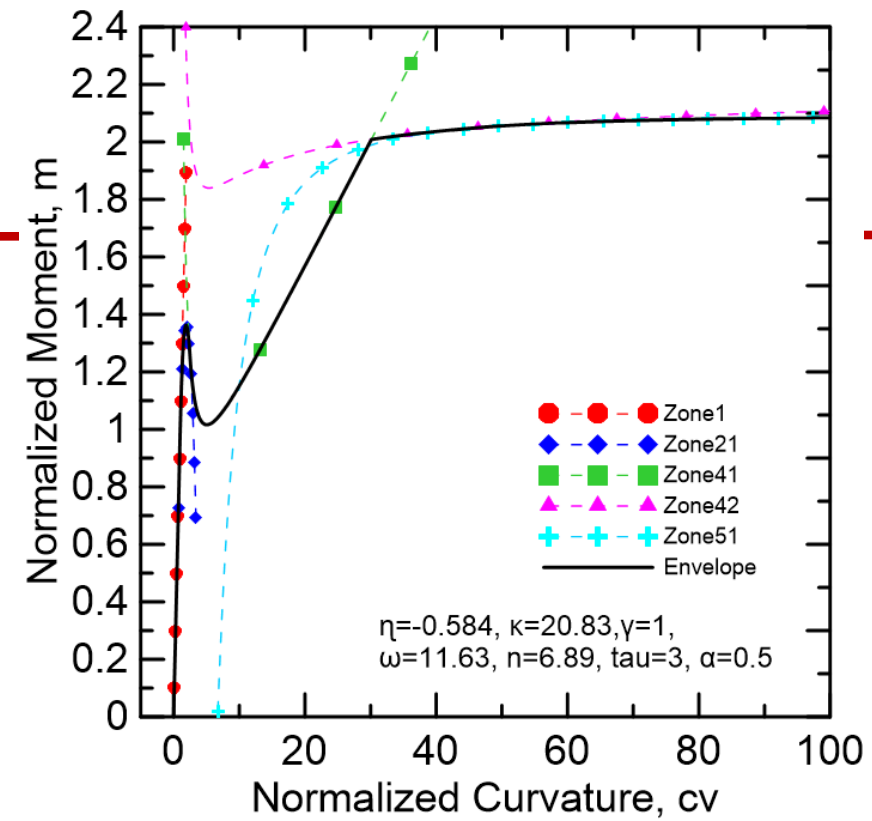
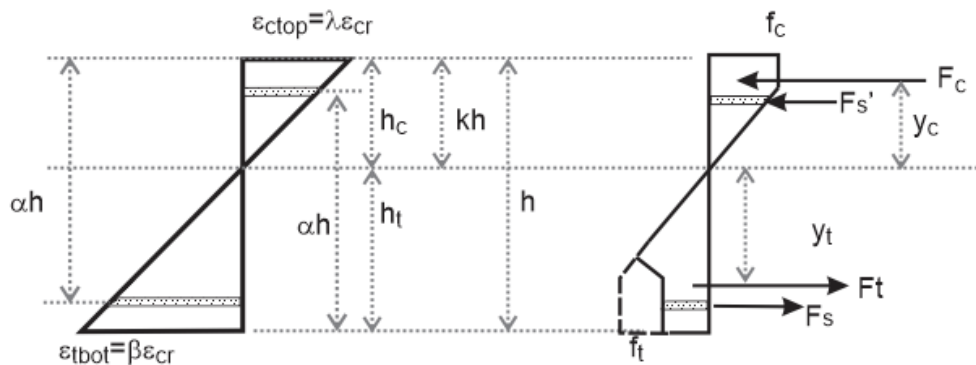
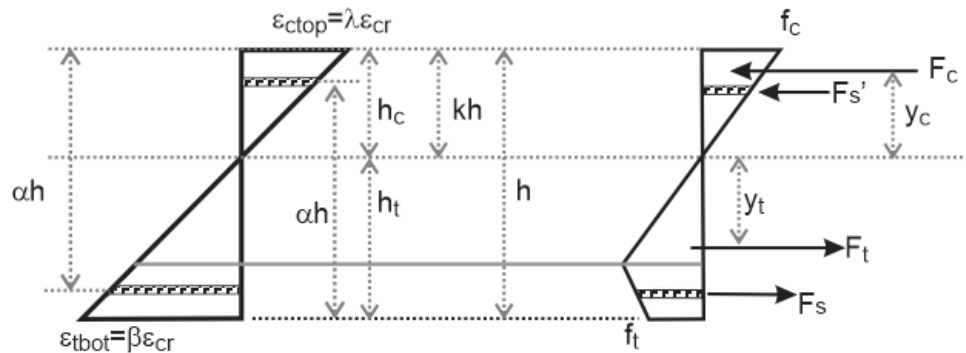
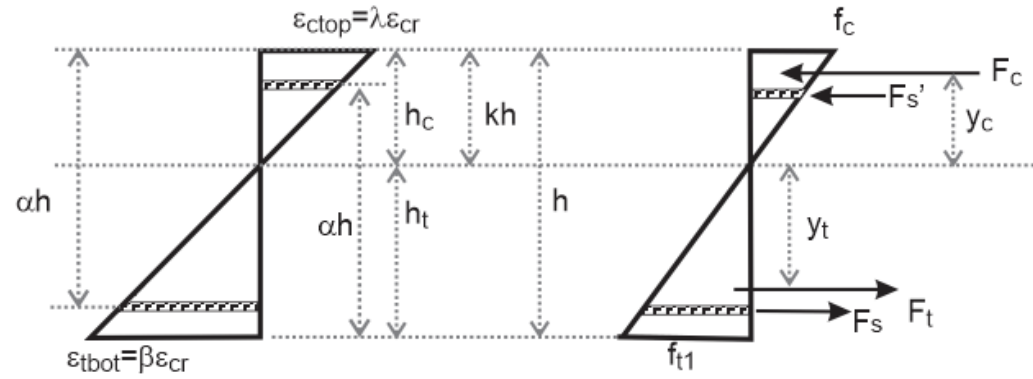
FRC Tension



Hybrid Section



Rebar

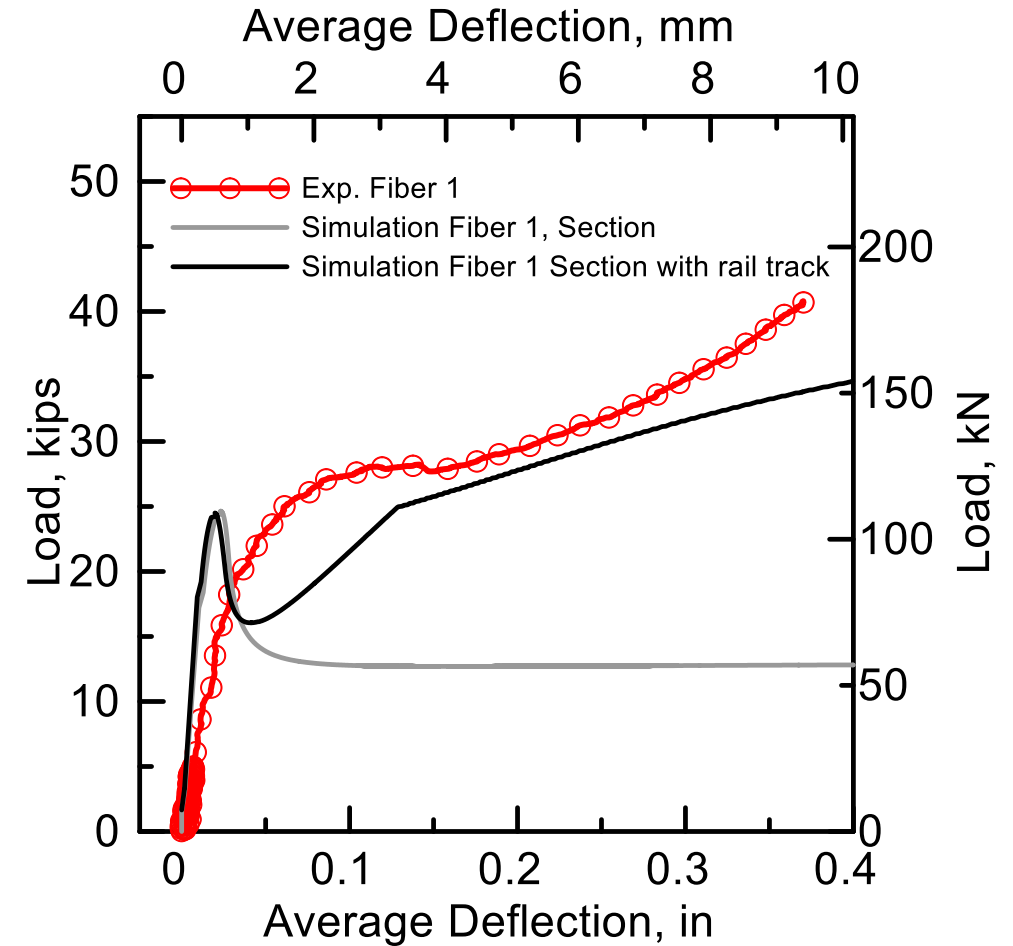
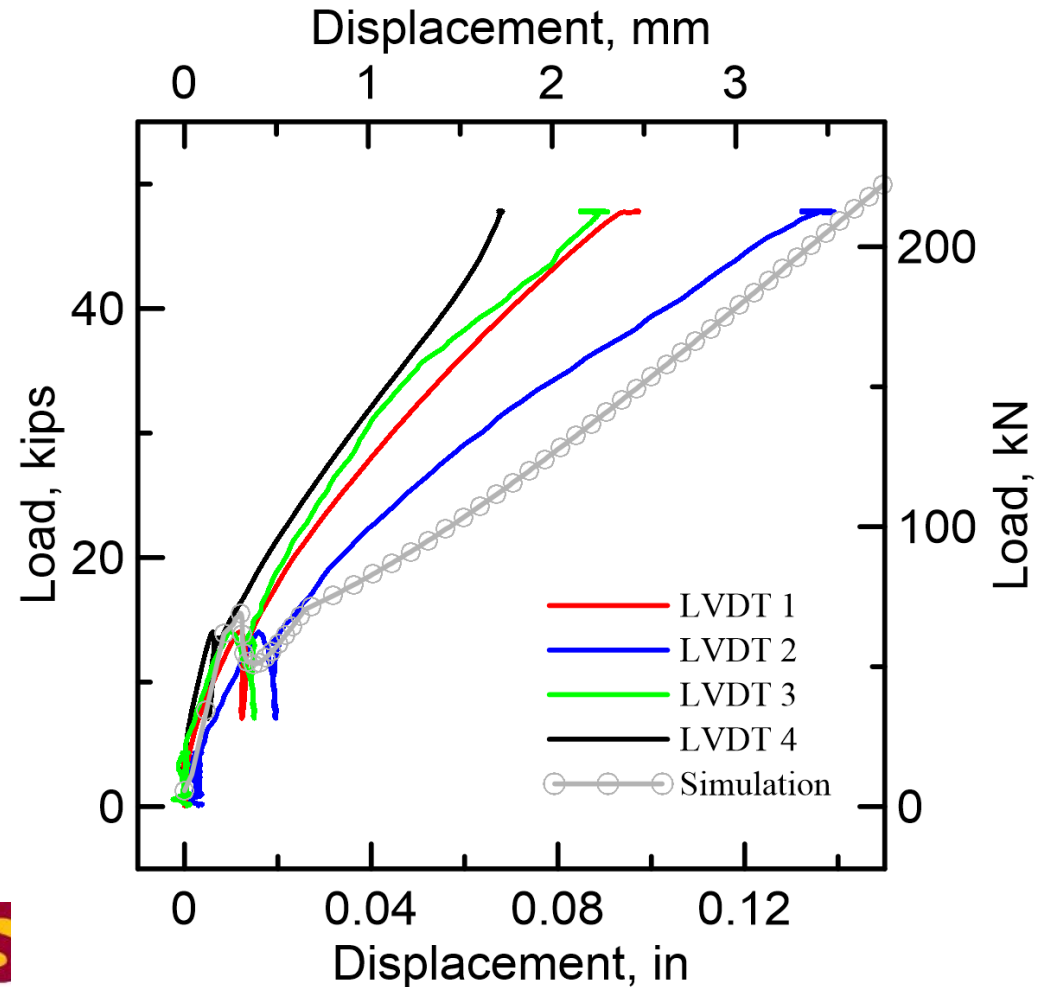


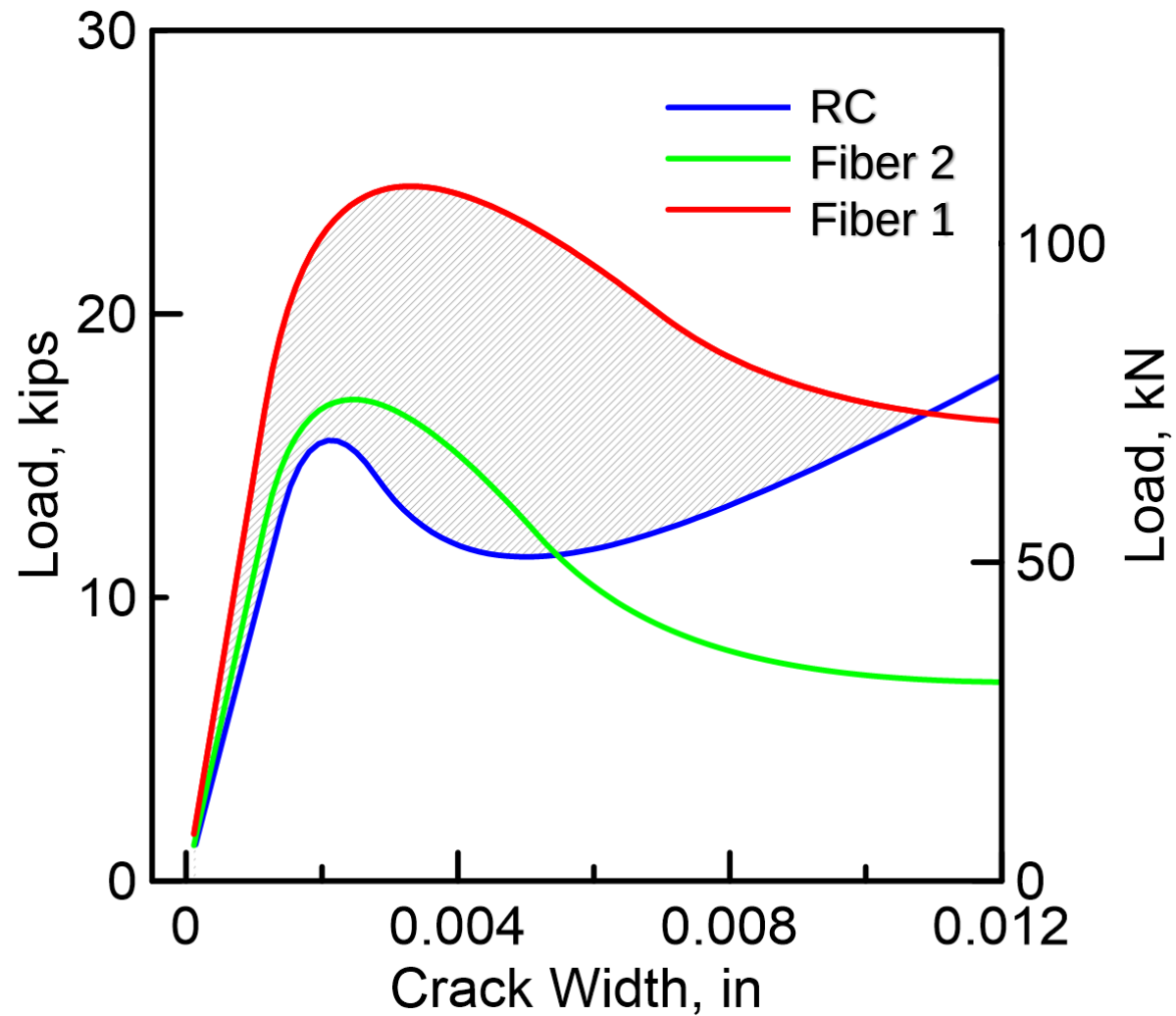
- Stress distribution corresponding to linear strain can use to generate moment capacity in beam section.

Rebar is assuming to perform the exact same strain with surrounding concrete matrix.

The figure show normalized moment capacity against curvature. Once M- ϕ response has known, Deflection can be calculated by using double integration along the beam length.

ASU Simulation Full-scale response to 3-point bending





Allowable crack width were recommended by standard guidelines of structure surrounding in environment and structure that exposed to the soil. To prevent excessive water leaks and reinforcement corrosion, crack width were limited below

JSCE, 2007 limit at 0.012 in

ACI 224 (2007) limit at 0.012 in

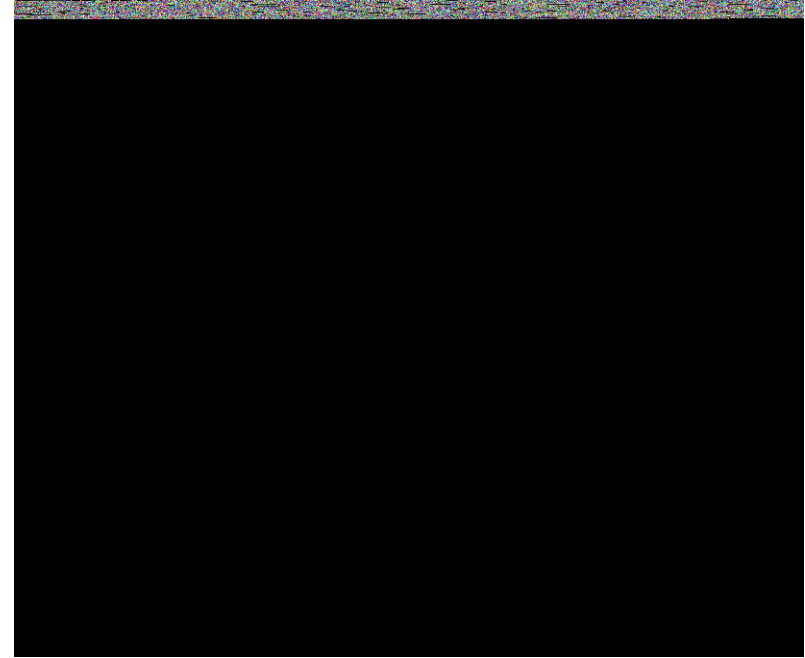
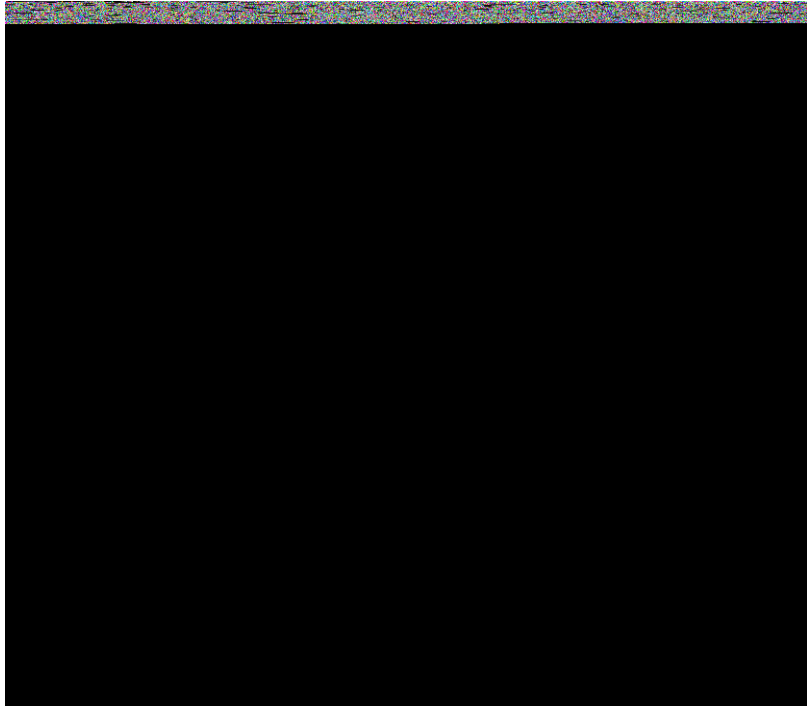
EN 1992-1,2004 limit at 0.012 in

Fib Model(2010) limit at 0.008 in

**** By concerning of serviceability design, it can be observed that the FRC section with d=12" present more energy absorption greater than conventional EC section with d=14.5"**

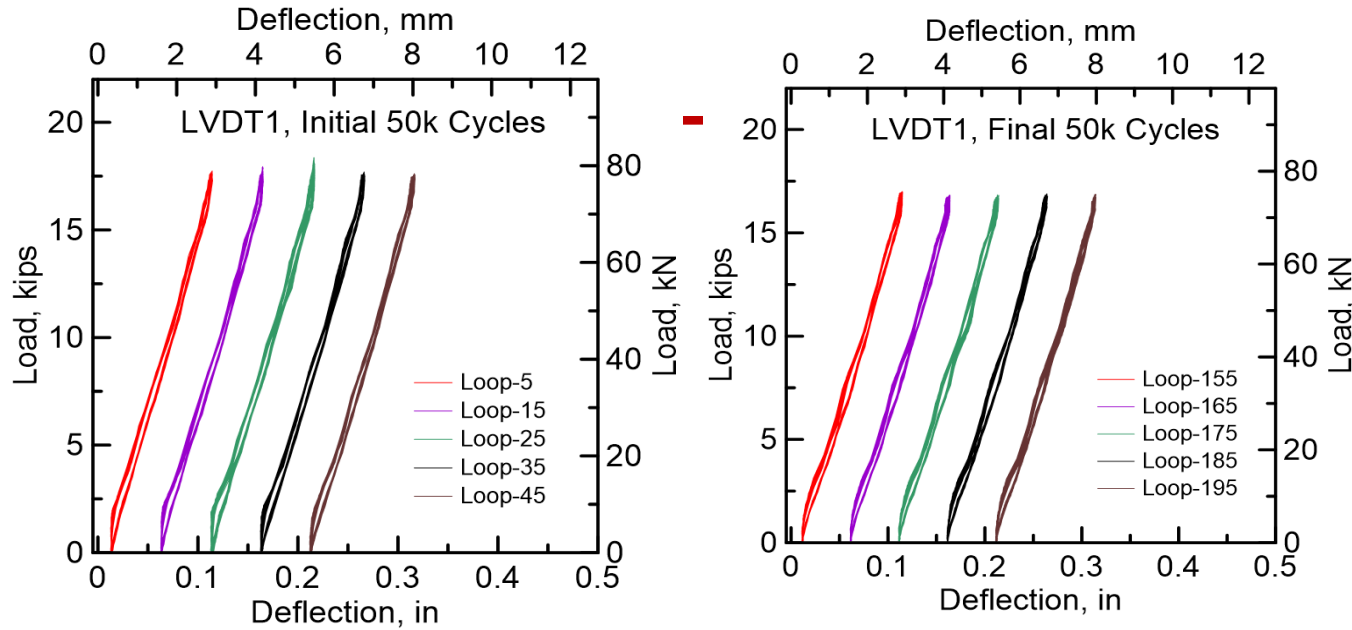
Conclusions

- The FRC specimens responses are commensurate with the amount of fibers used.
- The Fiber 1 specimens show a smooth transition from uncracked to cracked state. The increase in the post crack strength implies better performance that is attributed to the higher dosage of 65 lbs/yd³
- The capacity of the Fiber 2 decreases after the initial cracking at 22 kips after deformation is imposed, it recovers the load carrying capacity.
- The shape, length, and amount of fibers affect the final performance results. Design is therefore important.
- Table 1 shows the summary of all the monotonic testing.
- The deflection and deformation capacity of the FRC specimens is as much as three times the conventional design and thus noteworthy.

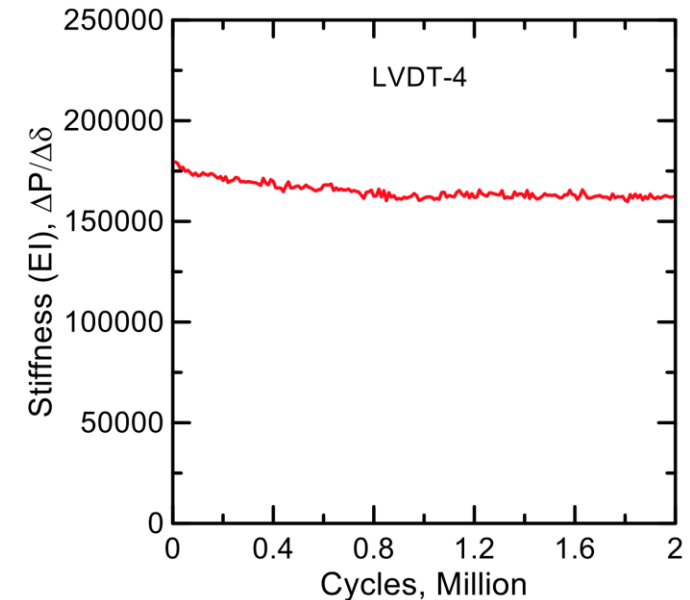
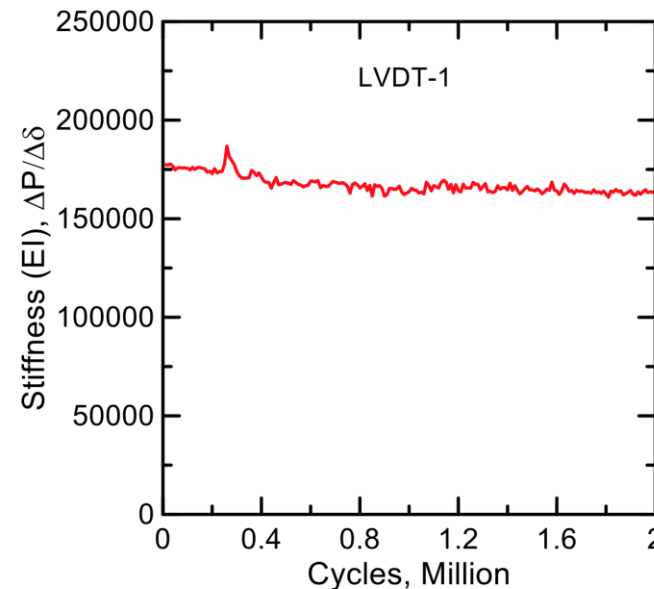


RC section was stiffer than FRC section. This is true when the bottom strain increasing, rebar contribute higher and higher load since modulus of rebar is around 7-8 times higher than FRC. Otherwise, if we considering the crack width limits, rebar will not be significant factor.

Fatigue Testing with Euclid Fibers (2 Million Cycles ,



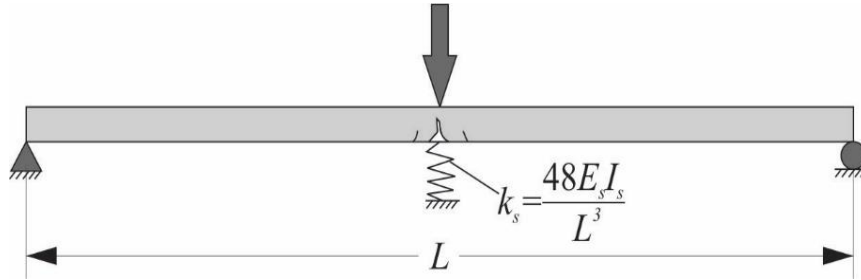
- specimen was subjected to 2 million cycles of fatigue representing 50-year service life under 17.5 kips per cycle loading which is 1.6-1.9 times the service load.
- Data Acquisition of 100 cycles per 10,000 cycles
- Average decrease in stiffness of 6.83% after 2 million cycles



Video Link of fatigue test:

https://youtu.be/z0Fi45NYh_k

Structural Calculation using 1D spring model



- Assumption Include Rail beam to be elastic to function as a spring of elastic modulus k

- Checking Elasticity of Steel Rail for Max loading 50 kips

$$M = \frac{PL}{4} = 1014000 \text{ lb} \cdot \text{in} \quad (P = 48000 \text{ lbs}, L = 84")$$

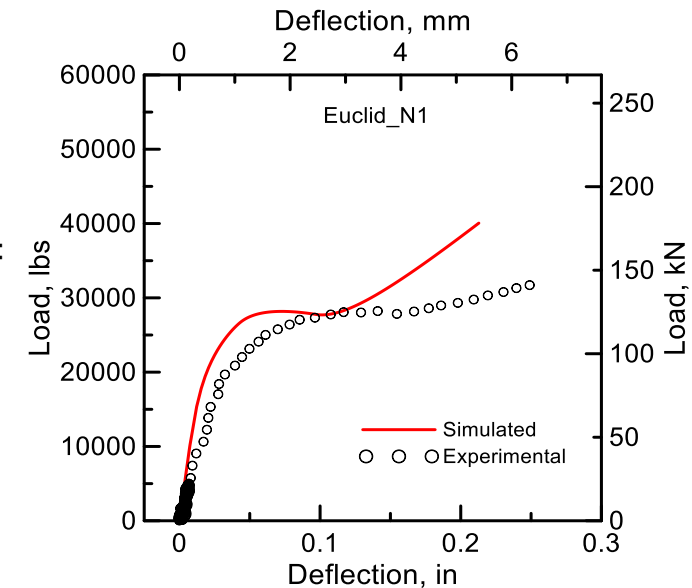
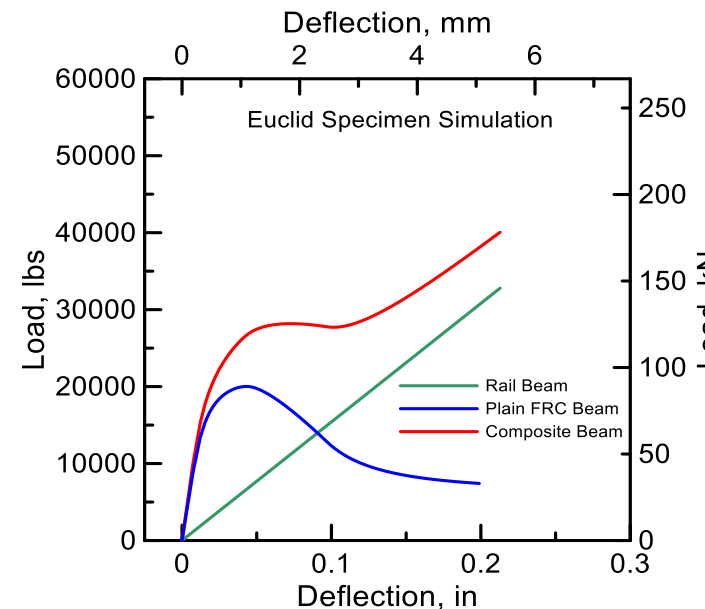
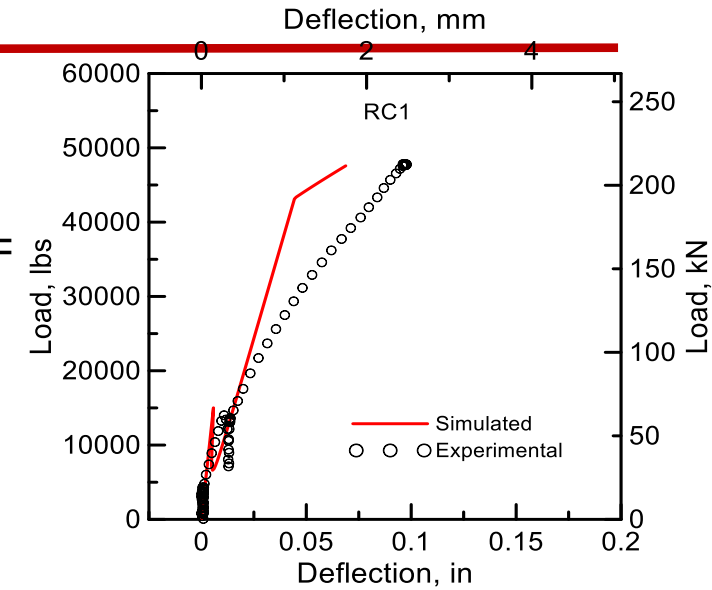
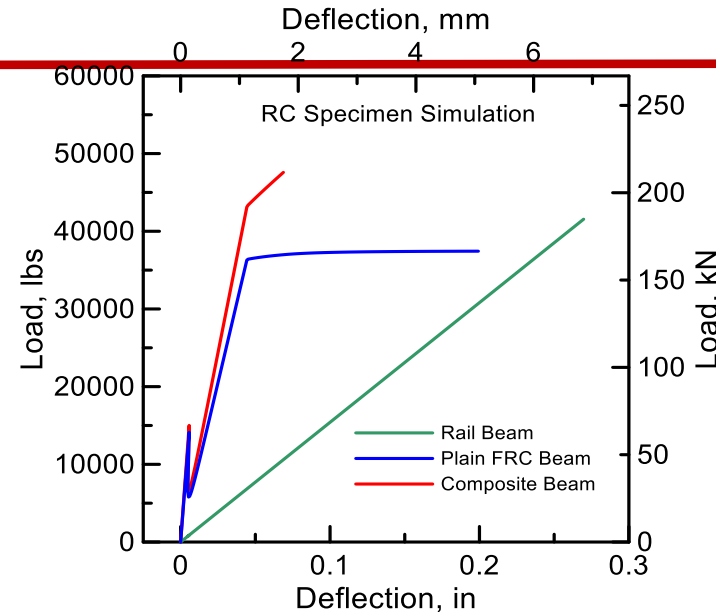
$$\sigma_{bottom} = \frac{My_{c,bottom}}{I} = 43667 \text{ psi} < 60000 \text{ psi}$$

$$\sigma_{top} = \frac{My_{c,top}}{I} = 57965 \text{ psi} < 60000 \text{ psi}$$

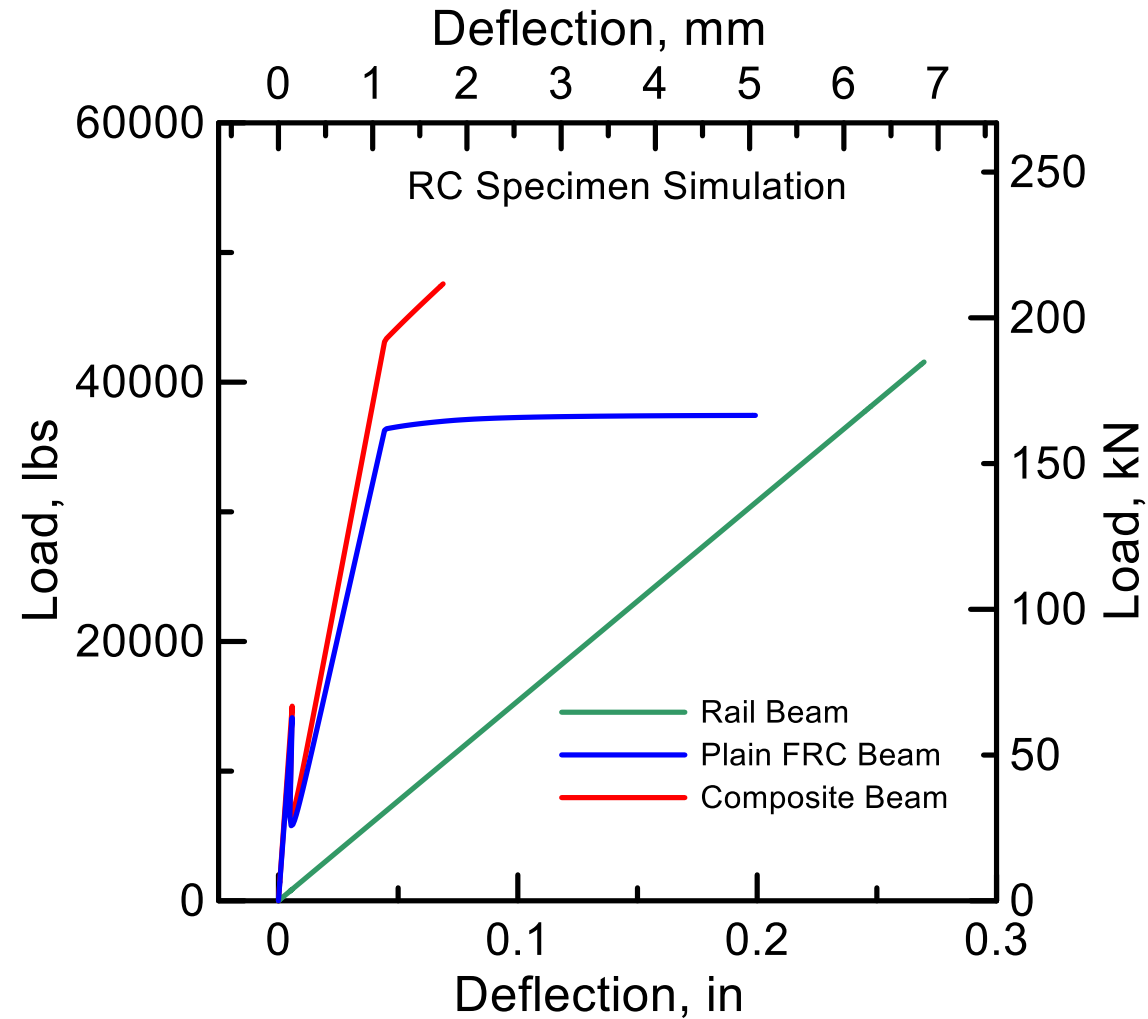
- Simulating plain FRC and RC Load-deflection response without rail beam based on back-calculated stress-strain parameters (ACI 544.4R approach)

Using Composite 2 beam equation

$$P(\delta) = P_s(\delta) + P_c(\delta)$$



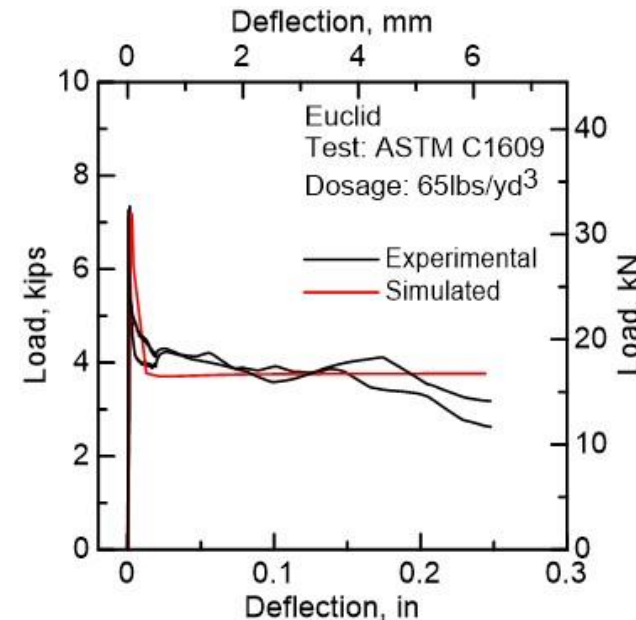
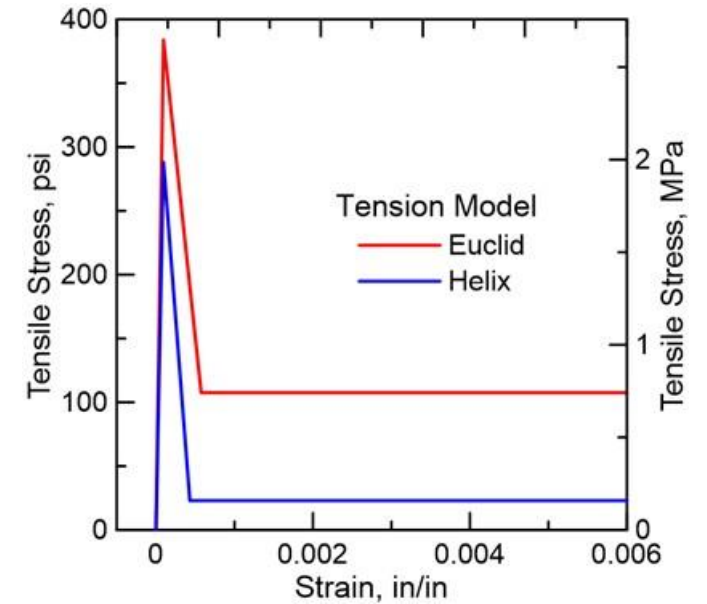
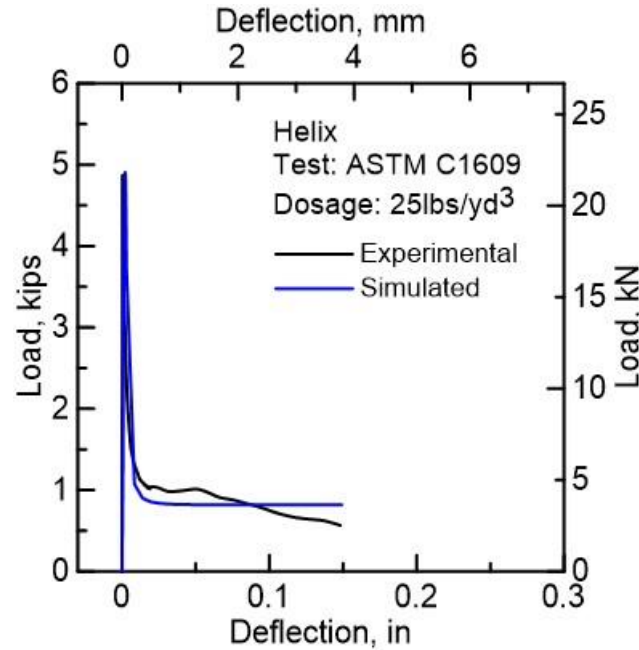
Composite Modeling of flexural loads



Structural Validation and Testing of FRC rail slabs

We need to say what the difference between the two mixtures are, how are these data used in the calculations

Extracting Material Properties of FRC



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